

Sharp Remainders and Exact Reductions in Polylogarithm Arithmetic Bridges

Log-gamma moments, Herglotz derivatives, and a Gaussian relation obstruction

A research continuation for the ProveIt Polylogarithms project
with OpenAI assistance

10 October 2026

Abstract

We continue the Polylogarithms programme with two quantitative asymptotic developments and exact algebraic support. For the classical factorial expansion of log-gamma moments, we identify the coefficient-generating inverse, prove its square-root singularity and complete late-coefficient expansion, and derive an asymptotic error equal to half the first omitted term at the least-term scale, including two correction terms. For normalized high derivatives of the Herglotz function, an exact two-term expression has an explicit remainder smaller than every power, with a stretched-exponential bound and an explicit oscillatory leading term. A finite rational algorithm gives all derivatives at positive rational arguments in Hurwitz-zeta, hence depth-one cyclotomic, coordinates. A sparse linear witness and all-weight dimension theorems explain a limitation of a specified depth-two relation presentation in the Gaussian S_4 problem. A new weight-seven harmonic-sum reduction is proposed with two independent high-precision checks and an exact rational residual bound of 10^{-260} . Finally, we correct the manuscript's description of the cubic log-gamma moment: a Tornheim–Witten derivative evaluation is classical, and we rederive it with convergence proofs and a reproducible evaluator. Every new claim is distinguished from classical input, formal algebra, numerical corroboration, and the conjectures left unresolved.

Contents

1	Scope, status, and principal results	3
1.1	What is proved here	3
1.2	Classical input and attribution	4
1.3	Conventions and verification levels	4
2	Log-gamma moments: from a classical expansion to a sharp error law	5
2.1	Inverse coordinates and residues	5
2.2	The singularity that controls late coefficients	7
2.3	A uniform Taylor-remainder lemma	8
2.4	Sharp optimal truncation	9
2.5	A convergent identity with an explicit tail bound	11
3	Rational Herglotz derivatives and an exponentially small oscillation	12
3.1	An explicit finite rational-jet formula	12
3.2	The complete algebraic large-order expansion	14
3.3	Exact arithmetic sectors and the leading oscillation	16

4	Cyclotomic double relations: exact identities and formal obstructions	19
4.1	Two exact weight-five reductions	20
4.2	An all-exponent mixed endpoint identity	20
4.3	Inverse-argument mixed values and a source correction	21
4.4	A precisely defined formal presentation	22
4.5	All-weight dimension theorems	23
4.6	A separating functional for the S_4 conjecture	24
4.7	A new weight-seven candidate	25
4.8	A rigorous rational certificate for the new conjecture	26
5	The cubic log-gamma moment: a classical evaluation and a source correction	27
5.1	A positive summand form and an explicit tail bound	29
5.2	Exponentially convergent numerical reproduction	29
5.3	A sign check for an auxiliary identity	31
6	Corrections, integration notes, and further research	31
6.1	Proposed corrections to the pinned manuscript	31
6.2	Integration and claim status	32
6.3	A proposed research programme	32
7	Reproducibility, numerical evidence, and limitations	34
7.1	Moment coefficients and least-term errors	34
7.2	Herglotz derivatives	35
7.3	Cyclotomic relation spaces and the conjecture certificate	35
7.4	Reproduction of the classical cubic evaluation	36
7.5	Running the package	36

1 Scope, status, and principal results

The ProveIt Polylogarithms manuscript [1] organizes a large collection of functional equations, special values, experimental reductions, and arithmetic bridges. This continuation concentrates on a connected part of that programme: logarithmic gamma moments, Herglotz derivatives, and depth-two cyclotomic polylogarithms. The common methodological question is how to turn a useful formal representation into a statement with controlled errors or a precisely specified algebraic meaning.

The source baseline is commit

78f9e5df05eb0abb16c89fb00b9e89403598bd17

of VladimirReshetnikov/ProveIt. We concentrate on Chapters 4, 7, 9, and 10 in `Analysis/Polylogarithms/docs/manuscript`. We also inspected the existing October continuations on alternating harmonic sums, corpus corrections, rational-grid distribution ranks, Stieltjes derivatives, and Herglotz obstructions. The accompanying source manifest records the examined files. This article proposes additions and corrections; it does not modify the upstream repository.

1.1 What is proved here

There are three principal developments.

First, the classical moment expansion

$$\frac{1}{n!} \int_0^1 \log^n \Gamma(x) dx \sim \sum_{k \geq 1} \frac{(-1)^{k-1} b_k}{k^n}$$

has coefficients that grow exponentially. We identify their precise growth and prove a sharp error law when the number of retained terms grows linearly with n . At the least-term scale the actual error is asymptotic to *one half of the first omitted term*; the article computes two correction terms. This is stronger than a fixed-length asymptotic expansion or a heuristic minimum-term estimate. The proof requires a local analytic continuation of an inverse gamma branch and a uniform estimate for a Taylor remainder across its circle of convergence.

Second, for

$$F(x) = \sum_{n \geq 1} \frac{\psi(nx) - \log(nx)}{n}, \quad D_r(x) = \frac{(-1)^{r+1} x^{r+1}}{r!} F^{(r)}(x),$$

we show that

$$D_r(x) = \zeta(2) + \frac{x}{r} (\log x - H_{r-1}) + E_r(x).$$

The displayed elementary expression contains every algebraic order of the large- r expansion. The remaining term admits a divisor-weighted Tricomi- U representation, an effective gamma-factor bound, and an explicit oscillatory asymptotic of size $r^{-3/4} e^{-2\sqrt{\pi r/x}}$. This yields the sharp exponential rate and infinitely many positive and negative remainders. Here the derivative order tends to infinity with x fixed; this is a different limit from the familiar large- x Bernoulli expansion.

Third, we determine all-weight dimensions of a *specified formal presentation* of imaginary depth-two values at levels 3 and 4, after quotienting by single products and the manuscript's g -family. The Hilbert series are

$$\frac{t^5}{(1-t^2)(1-t^6)} \quad \text{and} \quad \frac{t^3}{(1-t^2)^2},$$

respectively. A sparse functional separates the proposed Gaussian S_4 reduction from every row in that presentation. This identifies what the chosen relation family cannot establish. It is not a proof of linear independence of actual periods, and it is not a disproof of the conjecture.

These results are supported by exact rational row calculations, independent integral evaluations, precision increases, and explicit analytic tail bounds. Their proofs do not depend on the numerical checks.

1.2 Classical input and attribution

The moment expansion itself is due to Amdeberhan, Coffey, Espinosa, Koutschan, Manna, and Moll [2]. The square-root mechanism used for its late coefficients is classical analytic combinatorics [3]; its hypotheses and the constants are established here for the inverse relevant to the moments. The growing-index remainder analysis is the additional step.

Radchenko and Zagier [5] already state closure of all rational Herglotz derivatives in depth-one cyclotomic polylogarithms. We supply an explicit finite Bernoulli–Hurwitz coefficient formula and a regularization calculation, not a new claim of closure. Their arithmetic integral representations also underlie the divisor expansion used here. Large-parameter asymptotics of Tricomi’s function are classical [8, 7]; the effective sum over arithmetic sectors and its high-derivative consequences are developed in this article.

The cubic log-gamma moment already has a Tornheim–Witten derivative evaluation in Bailey–Borwein–Borwein [4]. We correct the manuscript’s open-problem wording and independently derive an equivalent formula, including convergence details and an evaluator. A reduction to a more restrictive prescribed basket of ordinary zeta derivatives remains a different question.

The all-depth odd-harmonic-sum reflection and several low-weight Gaussian reductions have already been proved in the repository’s continuations. We use them as established context. Likewise, a count of survivors in a relation matrix is always tied to the exact relations imposed. Zhao’s broader work on cyclotomic relations [9] provides a useful warning against identifying a restricted row space with the full algebra of periods.

1.3 Conventions and verification levels

Throughout, $\log$ is the real logarithm on positive real arguments; complex powers and the function U use principal branches unless otherwise specified. We write $H_n = \sum_{j=1}^n j^{-1}$, $H_0 = 0$, $\psi = \Gamma'/\Gamma$, and

$$\mathrm{Li}_{a,b}(u, v) = \sum_{m>n\geq 1} \frac{u^m v^n}{m^a n^b}.$$

At convergent boundary points this is the ordinary value. When a leading $\mathrm{Li}_1(1)$ is required in a relation, the regularization is specified explicitly. The Dirichlet beta and eta functions are

$$\beta(s) = \sum_{n\geq 0} \frac{(-1)^n}{(2n+1)^s}, \quad \eta(s) = \sum_{n\geq 1} \frac{(-1)^{n-1}}{n^s},$$

with $\eta(1) = \log 2$ and $G = \beta(2)$.

A theorem in this article has an analytic or exact algebraic proof. A formal dimension is the dimension of the defined presentation. An integer-relation candidate is a conjecture even when its residual is extraordinarily small. Most numerical diagnostics use ordinary high-precision arithmetic, so analytic truncation estimates are distinguished from machine-certified rounding enclosures. The S_6 residual certificate instead uses exact rational arithmetic with proved tails. No claim of a formal proof-assistant verification is made.

The novelty assessment is relative to the pinned manuscript, the inspected continuations, and the cited primary literature. The article does not claim an exhaustive priority search. All statements presented as theorems stand on their supplied proofs, independently of that bibliographic qualification.

2 Log-gamma moments: from a classical expansion to a sharp error law

For $n \geq 0$, write

$$M_n = \int_0^1 \log^n \Gamma(x) dx, \quad m_n = \frac{M_n}{n!}.$$

The convention $M_0 = 1$ is understood. The classical expansion of Amdeberhan et al. [2] is

$$m_n \sim \sum_{k \geq 1} \frac{(-1)^{k-1} b_k}{k^n}, \quad b_k = \frac{1}{k} [z^{k-1}] \Gamma(1-z)^k > 0. \quad (2.1)$$

Here and below an infinite asymptotic expansion means that its truncation index is fixed before $n \rightarrow \infty$. It does not assert convergence of the infinite sum for any fixed n . Our aim is to determine the growth of b_k , locate the least term, and prove the error when the truncation index grows with n .

2.1 Inverse coordinates and residues

Let g denote the inverse germ at zero of the entire reciprocal gamma function:

$$\frac{1}{\Gamma(g(t))} = t, \quad g(0) = 0, \quad g'(0) = 1.$$

For $0 \leq t \leq 1$, use the real continuation with $0 \leq g(t) \leq 1$. It exists uniquely: $\psi(x) < 0$ for $0 < x \leq 1$, because ψ is increasing and $\psi(1) = -\gamma < 0$. Thus $1/\Gamma(x)$ is strictly increasing from 0 to 1 on this interval. The inverse under discussion is not the increasing large-argument inverse of Γ .

Proposition 2.1 (Inverse and Mellin descriptions). *With $c_k = (-1)^{k-1} b_k$, the inverse germ satisfies*

$$g(t) = \sum_{k \geq 1} c_k t^k, \quad B(t) := -g(-t) = \sum_{k \geq 1} b_k t^k = t\Gamma(1 - B(t)). \quad (2.2)$$

For $n \geq 1$,

$$m_n = \frac{1}{(n-1)!} \int_0^\infty y^{n-1} g(e^{-y}) dy. \quad (2.3)$$

The function

$$\mathcal{M}(s) = \int_0^1 \Gamma(x)^s dx, \quad \Re s < 1,$$

continues meromorphically to $\mathbb{C}$, has only simple poles at the positive integers, and

$$\text{Res}_{s=k} \mathcal{M}(s) = -k c_k, \quad [s^n] \mathcal{M}(s) = m_n. \quad (2.4)$$

Every positive integer is an actual pole.

Proof. The inverse equation is $g = t\Gamma(1+g)$. Lagrange inversion gives $[t^k]g = k^{-1}[z^{k-1}]\Gamma(1+z)^k = c_k$. The expansion

$$\Phi(z) := \Gamma(1-z) = \exp\left(\gamma z + \sum_{j \geq 2} \frac{\zeta(j)}{j} z^j\right) \quad (2.5)$$

has strictly positive coefficients. This proves the positivity in (2.1) and the equation for B . The function $x \mapsto \log \Gamma(x)$ decreases from infinity to zero; the measure of the set where it exceeds y is $g(e^{-y})$. The layer-cake formula for a nonnegative function gives (2.3).

For the last assertion, set $H(x, s) = \Gamma(1+x)^s$, using its analytic logarithm for $|x| < 1$. Write $H(x, s) = \sum_{j \geq 0} a_j(s)x^j$, with each a_j entire in s . Choose $0 < d < 1$, split the integral at d , and subtract a finite Taylor polynomial on $[0, d]$. For every integer $J \geq 0$,

$$\mathcal{M}(s) = \sum_{j=0}^J \frac{a_j(s)d^{j+1-s}}{j+1-s} + \int_0^d x^{-s} \left(H(x, s) - \sum_{j=0}^J a_j(s)x^j \right) dx + \int_d^1 \Gamma(x)^s dx. \quad (2.6)$$

The remainder integral is holomorphic in $\Re s < J+2$, locally uniformly there. Taking J arbitrarily large proves continuation. At $s = k$, the residue is $-a_{k-1}(k) = -kc_k$, which is nonzero. Differentiation under the original integral near $s = 0$ proves the coefficient identity. $\square$

Subtracting the first K polar parts $kc_k/(k-s)$ from $\mathcal{M}$ leaves a function holomorphic in $|s| < K+1$. Cauchy's estimate, after subtracting one extra pole, consequently gives the rigorous fixed-length version

$$m_n = \sum_{k=1}^K \frac{c_k}{k^n} + \frac{c_{K+1}}{(K+1)^n} + O_{K,R}(R^{-n}), \quad K+1 < R < K+2. \quad (2.7)$$

This recovers the classical expansion by a meromorphic argument; it is not presented as a new asymptotic scale.

For exact coefficient generation, define $a_0(s) = 1$ and

$$a_j(s) = \frac{s}{j} \sum_{\ell=1}^j q_\ell a_{j-\ell}(s), \quad q_1 = \gamma, \quad q_\ell = \zeta(\ell) \quad (\ell \geq 2).$$

Then $b_k = a_{k-1}(k)/k$. This recurrence involves only rational arithmetic in the formal symbols $\gamma, \zeta(2), \dots$. For example,

$$\begin{aligned} b_1 &= 1, & b_2 &= \gamma, & b_3 &= \frac{3\gamma^2 + \zeta(2)}{2}, \\ b_4 &= \frac{8\gamma^3 + 6\gamma\zeta(2) + \zeta(3)}{3}, \\ b_5 &= \frac{125\gamma^4}{24} + \frac{25\gamma^2\zeta(2)}{4} + \frac{5\gamma\zeta(3)}{3} + \frac{5\zeta(2)^2}{8} + \frac{\zeta(4)}{4}. \end{aligned}$$

These homogeneous polynomials have weight $k-1$ under the formal assignments $\text{wt}(\gamma) = 1$ and $\text{wt}(\zeta(j)) = j$. No independence assumption is needed to generate or verify them.

2.2 The singularity that controls late coefficients

Theorem 2.2 (Late coefficients). *There is a unique $\tau \in (0, 1)$ satisfying $\psi(-\tau) = 0$. Define*

$$\rho = \frac{\tau}{\Gamma(1-\tau)}, \quad \Lambda = -\log \rho, \quad C = \frac{1}{\sqrt{2\pi\psi'(-\tau)}}. \quad (2.8)$$

Then B has radius of convergence ρ , its only singularity on that circle is ρ , and it has a square-root expansion there. In particular,

$$b_k = C\rho^{-k}k^{-3/2} \left(1 + \frac{d_1}{k} + O(k^{-2}) \right). \quad (2.9)$$

A complete expansion in powers of $1/k$ exists. If $\kappa_j = ((z\partial_z)^j \log \Phi(z))|_{z=\tau}$, so that $\kappa_1 = 1$, then

$$d_1 = -\frac{1}{2\kappa_2} + \frac{\kappa_3}{2\kappa_2^2} + \frac{\kappa_4}{8\kappa_2^2} - \frac{5\kappa_3^2}{24\kappa_2^3}. \quad (2.10)$$

Equivalently, with $P = \psi'(-\tau)$, $Q = \psi''(-\tau)$, and $R = \psi'''(-\tau)$,

$$d_1 = \frac{R}{8P^2} - \frac{5Q^2}{24P^3}. \quad (2.11)$$

Proof. The characteristic equation for $B = t\Phi(B)$ is $\tau\Phi'(\tau) = \Phi(\tau)$. Since $\Phi'/\Phi = -\psi(1-z)$, recurrence of the digamma function turns this into $\psi(-\tau) = 0$. Alternatively $z\Phi'(z)/\Phi(z) = \gamma z + \sum_{j \geq 2} \zeta(j)z^j$ is strictly increasing from 0 to infinity on $(0, 1)$, proving existence and uniqueness. The function $z/\Phi(z)$ increases up to τ and then decreases.

The power series solution has nonnegative coefficients and reaches $B(\rho) = \tau$. One elementary justification is to iterate $B_{j+1} = t\Phi(B_j)$, starting with $B_0 = 0$, for $0 \leq t < \rho$; monotonicity and the smaller real solution bound the iterates. Monotone convergence and inversion give the limit at ρ . At the characteristic point,

$$\Phi''(\tau)/\Phi(\tau) = \psi'(1-\tau) + \psi(1-\tau)^2 = \psi'(1-\tau) + \tau^{-2} = \psi'(-\tau) > 0.$$

Local inversion therefore gives

$$B(t) = \tau - \alpha\sqrt{1-t/\rho} + O(1-t/\rho), \quad \alpha = \sqrt{\frac{2}{\psi'(-\tau)}}. \quad (2.12)$$

There is a convergent Puiseux expansion, with integral and half-integral powers.

For completeness, there are no other dominant singularities. Absolute convergence at $|t| = \rho$ follows from $B(\rho) = \tau$. If $t \neq \rho$ lies on that circle, the positive coefficients of t and t^2 give $|B(t)| < \tau$. Hence

$$|t\Phi'(B(t))| \leq \rho\Phi'(|B(t)|) < \rho\Phi'(\tau) = 1.$$

The analytic implicit-function theorem continues B across every such point. Compactness, together with local square-root inversion at ρ , supplies a slit disk $|t| < \rho + \epsilon$ with the sole cut issuing from ρ . This is the smooth implicit-function mechanism of analytic combinatorics [3], with its hypotheses verified here. Transfer of $(1-t/\rho)^{1/2}$ gives $C = \alpha/(2\sqrt{\pi})$ and the full expansion.

We also give a direct computation of the first correction. Lagrange's coefficient integral on $|z| = \tau$ is

$$b_k = \frac{\tau\rho^{-k}}{2\pi k} \int_{-\pi}^{\pi} \exp(k\{\log \Phi(\tau e^{i\theta}) - \log \Phi(\tau) - i\theta\}) e^{i\theta} d\theta.$$

Positivity and aperiodicity make the modulus strictly smaller away from $\theta = 0$. At zero the exponent is $-\kappa_2\theta^2/2 - i\kappa_3\theta^3/6 + \kappa_4\theta^4/24 + \dots$. After $\theta = u/\sqrt{k}$, Gaussian integration gives respectively $-1/(2\kappa_2)$, $\kappa_3/(2\kappa_2^2)$, $\kappa_4/(8\kappa_2^2)$, and $-5\kappa_3^2/(24\kappa_2^3)$. Also $\kappa_2 = \tau^2\psi'(-\tau)$, so the leading constant agrees with (2.8). Taylor expansion to any fixed order, with an exponentially small outer arc, proves the complete saddle expansion. $\square$

Numerical values, used only as orientation, are

$$\begin{aligned}\tau &= 0.5040830082644554092582693045333024989\dots, \\ \rho &= 0.2821158090062980421537966006204637282\dots, \\ \Lambda &= 1.2654376221108656133820359111654739922\dots, \\ C &= 0.1334277613094452848960205705394178243\dots, \\ d_1 &= 0.3019628287812052411245866827164781138\dots\end{aligned}$$

Corollary 2.3 (Divergence and least terms). *For every fixed n , the terms of the infinite series in (2.1) eventually grow in magnitude and do not tend to zero. Its smallest terms occur at*

$$k = \frac{n + 3/2}{\Lambda} + O(1).$$

For any integer N with $N\Lambda - n$ bounded,

$$\frac{b_N}{N^n} = C \left(\frac{e\Lambda}{n + 3/2} \right)^{n+3/2} (1 + O(n^{-1})). \quad (2.13)$$

Proof. Equation (2.9) gives exponential growth ρ^{-k} ; $\rho < 1$, for example because $\Phi(\tau) > 1$ and $\tau < 1$. The full coefficient expansion gives

$$\log \frac{b_{k+1}/(k+1)^n}{b_k/k^n} = \Lambda - (n + 3/2) \log(1 + 1/k) + O(k^{-2}).$$

This changes sign in a bounded interval around the displayed value. No fixed initial index can be a global minimum for large n , and the same ratio excludes the other ranges. Finally, the function $k\Lambda - (n + 3/2) \log k$ has a stationary point at $(n + 3/2)/\Lambda$; a bounded shift changes it by $O(n^{-1})$. $\square$

2.3 A uniform Taylor-remainder lemma

The next statement is the key passage from fixed-order asymptotics to optimal truncation. It explicitly crosses the radius of convergence on the positive real axis; merely summing an infinite Taylor tail there would be invalid.

Lemma 2.4 (Uniform inverse remainder). *For some $\delta > 0$, uniformly for $t \in [\rho e^{-\delta}, \rho e^{\delta}]$, put $r = t/\rho$. Then*

$$\frac{g(t) - \sum_{k=1}^{N-1} c_k t^k}{c_N t^N} = \frac{1}{1+r} + \frac{3r}{2N(1+r)^2} + O(N^{-2}). \quad (2.14)$$

On each of the remaining real intervals $0 < t \leq \rho e^{-\delta}$ and $\rho e^{\delta} \leq t \leq 1$, the same normalized remainder is bounded uniformly in N sufficiently large.

Proof. The preceding theorem gives a slit disk of continuation for g , with its only singularity at $-\rho$. Choose its outer radius $R > \rho$, and then $\delta > 0$ so small that $\rho e^\delta < R$ and $\rho e^\delta < 1$. Cauchy's integral formula for $g(t)$, less its first $N - 1$ Taylor terms, has kernel $t^N/(z^N(z - t))$. Deform its contour to the two banks of the negative cut and an outer contour. The coefficient c_N uses kernel z^{-N-1} . On the cut $z = -s$, the ratio of these kernels, after removing t^N , is

$$\frac{z}{z - t} = \frac{s}{s + t}.$$

The jump of g at $s = \rho$ is a nonzero constant times $(s - \rho)^{1/2}(1 + O(s - \rho))$, with a complete expansion. The coefficient integral is thus a Watson-type endpoint integral whose normalized variable $u = (s - \rho)/\rho$ has

$$\langle u \rangle = \frac{3}{2N} + O(N^{-2}), \quad \langle u^2 \rangle = O(N^{-2}).$$

These formulas follow directly by comparing $\int_0^\epsilon u^{j+1/2}(1+u)^{-N-1} du$ with its beta integral extended to infinity. Terms outside a fixed small neighborhood of zero are exponentially smaller. Expanding

$$\frac{s}{s + t} = \frac{1}{1 + r} + \frac{r}{(1 + r)^2} u + O(u^2)$$

proves (2.14). The outer contour contributes $O(R^{-N})$ to the coefficient and $O((t/R)^N)$ to the remainder; after normalization these errors are exponentially small uniformly on the stated compact t -interval. This contour argument defines and estimates the actual analytic remainder on both sides of $t = \rho$.

It remains to bound the two outer real intervals. The estimate $b_k \asymp \rho^{-k} k^{-3/2}$ implies that, below $\rho e^{-\delta}$, the absolutely convergent tail is bounded by a constant times $b_N t^N$. Above ρe^δ , use

$$\sum_{k < N} \frac{b_k t^k}{b_N t^N} \ll \sum_{k < N} (N/k)^{3/2} (\rho/t)^{N-k}.$$

Splitting at $k = N/2$ bounds this uniformly by a geometric series plus an exponentially small term times a polynomial in N . Furthermore $0 \leq g(t) \leq 1$, whereas $b_N t^N$ grows exponentially uniformly on this interval. This proves the last claim. $\square$

2.4 Sharp optimal truncation

Theorem 2.5 (Half the first omitted term). *Let n, N tend to infinity through positive integers with $\eta = N\Lambda - n$ in a fixed bounded interval. Then*

$$m_n - \sum_{k=1}^{N-1} \frac{(-1)^{k-1} b_k}{k^n} = \frac{(-1)^{N-1} b_N}{N^n} \left(\frac{1}{2} + \frac{3 - 2\eta}{8N} + O(N^{-2}) \right). \quad (2.15)$$

The O -constant is uniform for bounded η . The ratio in parentheses has a complete asymptotic expansion in $1/N$, with coefficients computable from the local inverse expansion and gamma moments. In particular, taking the nearest integer to $(n + 3/2)/\Lambda$ gives a rigorously justified least-term approximation. The error eventually has the sign of the first omitted term and is asymptotic to one half of that term.

Proof. Subtract the finite sum under (2.3), using

$$\frac{1}{(n-1)!} \int_0^\infty y^{n-1} e^{-ky} dy = k^{-n}.$$

With Y having the gamma density $N^n y^{n-1} e^{-Ny} / (n-1)!$, the error divided by c_N / N^n is the expectation of the normalized inverse remainder at $t = e^{-Y}$.

The gamma distribution concentrates at Λ :

$$\mathbb{E}Y = \frac{n}{N} = \Lambda - \frac{\eta}{N}, \quad \text{Var } Y = \frac{n}{N^2}.$$

For every fixed small $\delta > 0$, its probability outside $[\Lambda - \delta, \Lambda + \delta]$ is $O(e^{-a_\delta N})$. This follows from its moment-generating function $(1 - u/N)^{-n}$ and Chernoff's inequality. The uniform outer bounds in Lemma 2.4 therefore make the outer contribution exponentially small.

On the middle interval, the leading kernel is

$$L(Y - \Lambda) = \frac{1}{1 + e^{\Lambda - Y}}, \quad L(v) = \frac{1}{2} + \frac{v}{4} - \frac{v^3}{48} + O(v^4).$$

Writing $V = Y - \Lambda$, its centered gamma moments give

$$\mathbb{E}V = -\frac{\eta}{N}, \quad \mathbb{E}V^3 = O(N^{-2}), \quad \mathbb{E}V^4 = O(N^{-2}),$$

uniformly for bounded η . Thus $\mathbb{E}L(V) = 1/2 - \eta/(4N) + O(N^{-2})$. It is essential here to use the signed third moment; an absolute cubic bound alone would lose half a power. The correction kernel in (2.14) is

$$\frac{3e^{-V}}{2N(1 + e^{-V})^2}.$$

Its value at zero is $3/(8N)$, and its first derivative vanishes there, so its expectation is $3/(8N) + O(N^{-2})$. Adding these terms proves (2.15). The contour calculation and the Taylor/gamma-moment calculation can both be continued to arbitrary fixed order. Gamma central moments are polynomials in n and N^{-1} ; their parity and degree give integer powers of N^{-1} . $\square$

The theorem strengthens the classical expansion in two ways. It allows N to grow linearly with n , and it estimates the actual error after truncation. The least-term estimate alone would not justify either conclusion. No global sign assertion for every n, N is needed.

Corollary 2.6 (Second correction). *Under the hypotheses of Theorem 2.5, the normalized error has the more precise expansion*

$$\begin{aligned} \frac{m_n - \sum_{k < N} c_k / k^n}{c_N / N^n} &= \frac{1}{2} + \frac{3 - 2\eta}{8N} \\ &\quad + \frac{1}{N^2} \left(\frac{d_1}{4} + \frac{\Lambda(6\eta - 13)}{96} \right) + O(N^{-3}). \end{aligned} \quad (2.16)$$

Proof. Here are the additional local coefficients, so that the assertion is independently reproducible. Let $v = B - \tau$, and use P, Q, R from (2.11). The characteristic inverse satisfies

$$\log \frac{B/\Gamma(1-B)}{\rho} = -\frac{P}{2}v^2 + \frac{Q}{6}v^3 - \frac{R}{24}v^4 + O(v^5).$$

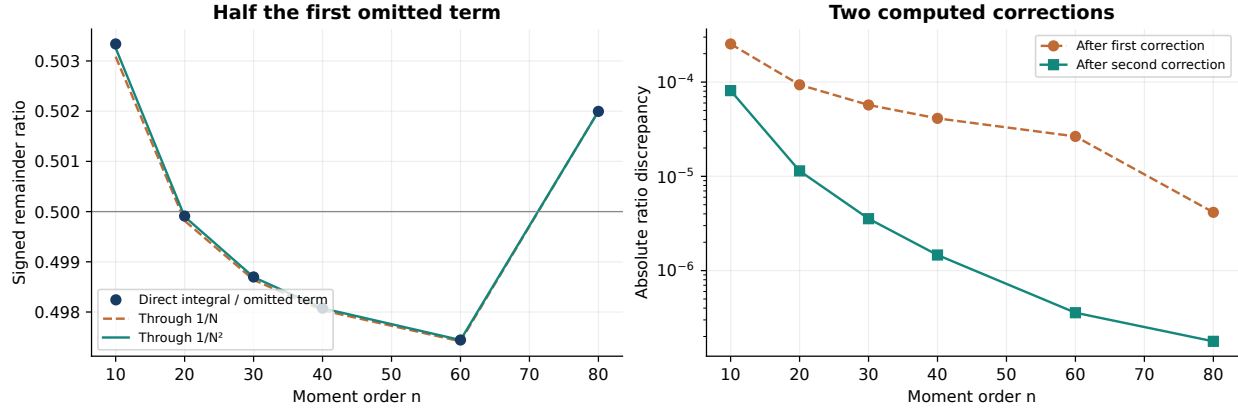


Figure 1: The signed error divided by the first omitted term, at a truncation near $(n + 3/2)/\Lambda$. The first-correction formula tracks the small changes caused by rounding the truncation index. The plot is numerical corroboration; Theorem 2.5 supplies the proof.

Putting $w = \sqrt{1 - t/\rho}$ and $\alpha = \sqrt{2/P}$, inversion gives

$$B(t) = \tau - \alpha w + \frac{Q}{3P^2}w^2 + \mu w^3 + O(w^4), \quad D := \frac{\mu}{\alpha} = -\frac{1}{4} + \frac{R}{12P^2} - \frac{5Q^2}{36P^3}.$$

Thus the positive cut density for B is $(\alpha/\pi)u^{1/2}(1 + Du + O(u^2))$, where $u = t/\rho - 1$. Transfer gives $d_1 = 3/8 + 3D/2$, proving (2.11) as well. The normalized endpoint moments are

$$\langle u \rangle = \frac{3}{2N} + \frac{9/4 + 3D/2}{N^2} + O(N^{-3}), \quad \langle u^2 \rangle = \frac{15}{4N^2} + O(N^{-3}), \quad \langle u^3 \rangle = O(N^{-3}).$$

They follow from beta integrals and the displayed cut density. The coefficient of N^{-2} in the inverse remainder at $r = t/\rho$ is therefore

$$K_2(r) = \frac{r}{(1+r)^2} \left(\frac{9}{4} + \frac{3D}{2} \right) - \frac{15r}{4(1+r)^3}, \quad K_2(1) = d_1/4.$$

For the gamma average, put $K_1(v) = 3e^{-v}/[2(1 + e^{-v})^2]$ and $V = Y - \Lambda$. Taylor expansion through degree five, using $\mathbb{E}|V|^6 = O(N^{-3})$, gives

$$\mathbb{E}L(V) = \frac{1}{2} - \frac{\eta}{4N} + \frac{\Lambda(3\eta - 2)}{48N^2} + O(N^{-3}), \quad \mathbb{E}K_1(V) = \frac{3}{8} - \frac{3\Lambda}{32N} + O(N^{-2}).$$

For example $\mathbb{E}V^3 = \Lambda(2 - 3\eta)N^{-2} + O(N^{-3})$; the signed fifth moment is $O(N^{-3})$. Also $\mathbb{E}K_2(e^{-V}) = K_2(1) + O(N^{-1})$. The uniform outer bound again makes all outer contributions exponentially small. Combining the three terms proves (2.16). $\square$

2.5 A convergent identity with an explicit tail bound

An asymptotic series should not be used as an infinite exact sum. The same inverse coordinate gives an exact convergent alternative.

Proposition 2.7 (Cutoff identity). *For $n \geq 1$ and $Y > \Lambda$,*

$$m_n = \frac{1}{(n-1)!} \int_0^Y y^{n-1} g(e^{-y}) dy + \sum_{k \geq 1} \frac{c_k}{k^n} Q(n, kY), \quad Q(n, z) = e^{-z} \sum_{j=0}^{n-1} \frac{z^j}{j!}. \quad (2.17)$$

The sum is absolutely convergent. If $Y > \log 4$, the absolute tail after $k = K$ is at most

$$\frac{(4e^{-Y})^{K+1}}{2(1-4e^{-Y})} \sum_{j=0}^{n-1} \frac{Y^j}{j!(K+1)^{n-j}}. \quad (2.18)$$

Proof. Split (2.3) at Y , substitute the convergent inverse series on the tail, and integrate term by term. The coefficient growth theorem proves absolute convergence. For the elementary explicit bound, note that

$$\frac{1/2}{\Phi(1/2)} = \frac{1}{2\sqrt{\pi}} > \frac{1}{4}.$$

The smaller solution to $B = t\Phi(B)$ at $t = 1/4$ is therefore less than $1/2$; equivalently $B(1/4) < 1/2$. Positivity gives $b_k \leq 4^k/2$. For $k > K$ and $j \leq n-1$, $k^{j-n} \leq (K+1)^{j-n}$. The geometric tail now gives (2.18). $\square$

The numerical implementation uses ordinary high-precision arithmetic and explicitly distinguishes analytic truncation bounds from interval rounding bounds. Formula (2.17) is also suitable for a future ball-arithmetic evaluator: the finite inverse integral and all constant evaluations can be enclosed separately.

3 Rational Herglotz derivatives and an exponentially small oscillation

Let

$$F(x) = \sum_{n \geq 1} \frac{\psi(nx) - \log(nx)}{n}, \quad D_r(x) = \frac{(-1)^{r+1} x^{r+1}}{r!} F^{(r)}(x) \quad (r \geq 1, x > 0).$$

The all-orders assertion that rational jets reduce to depth-one cyclotomic polylogarithms is already stated by Radchenko–Zagier [5], immediately after Proposition 4 of their preprint, p. 9. It must not be presented as a new closure theorem. The explicit Bernoulli coefficient formula below gives a constructive version; the large-order estimate and its effective remainder bound are derived independently here. The limit considered below is derivative order tending to infinity at a fixed positive argument.

3.1 An explicit finite rational-jet formula

Write $B_j(X)$ for the Bernoulli polynomials with $B_1(X) = X - 1/2$. For positive integers p, q (coprimality is unnecessary for this formula), put

$$c_{a,b} = \frac{a}{q} + \frac{b}{p}, \quad d_{r,j}(a,b) = \binom{r}{j} B_{r-j}(1 - b/p) - \mathbf{1}_{j=0} B_r(a/q),$$

where $1 \leq a \leq q$, $0 \leq b < p$, and $0 \leq j < r$.

Theorem 3.1 (Finite Bernoulli–Hurwitz evaluator). *For $r \geq 1$,*

$$F^{(r)}(p/q) = \frac{(-1)^{r+1}(r-1)!q^{r-1}}{p^{r+1}} \left\{ \sum_{a=1}^q \sum_{b=0}^{p-1} \left[-\psi(c_{a,b}) + \sum_{j=0}^{r-1} d_{r,j}(a,b) \zeta(r+1-j, c_{a,b}) \right] + pq \left(\frac{1}{r} - \gamma - \log q \right) \right\}.$$

In particular every Hurwitz zeta value in the formula has integer order between 2 and $r+1$; no spectral derivatives occur.

Proof. Use the classical integral representation (Radchenko–Zagier, eq. (6))

$$F^{(r)}(x) = (-1)^{r+1} \int_0^\infty t^r \operatorname{Li}_{1-r}(e^{-xt}) \left(\frac{1}{1-e^{-t}} - \frac{1}{t} \right) dt.$$

After $t = qu$ introduce, initially for $\Re s > 0$,

$$I(s) = \int_0^\infty u^{r+s} \operatorname{Li}_{1-r}(e^{-pu}) \left(\frac{1}{1-e^{-qu}} - \frac{1}{qu} \right) du.$$

The combined integral is holomorphic on $\Re s > -1$ since its integrand is $O(u^{\Re s})$ at zero. In $\Re s > 0$ the two terms can be integrated separately, giving

$$I(s) = \Gamma(r+1+s) \sum_{n \geq 1, m \geq 0} \frac{n^{r-1}}{(pn+qm)^{r+1+s}} - \frac{\Gamma(r+s)}{qp^{r+s}} \zeta(1+s).$$

Set $n = qk + a$, $m = p\ell + b$, and $h = k + \ell$. Faulhaber’s identity gives

$$\sum_{k=0}^h (qk+a)^{r-1} = \frac{q^{r-1}}{r} [B_r(h+1+a/q) - B_r(a/q)].$$

Writing $v = h + c_{a,b}$ expands the bracket as $\sum_{j=0}^r d_{r,j}(a,b) v^j$, with $d_{r,r} = 1$. Consequently

$$\sum_{n \geq 1, m \geq 0} \frac{n^{r-1}}{(pn+qm)^{r+1+s}} = \frac{q^{r-1}}{r(pq)^{r+1+s}} \sum_{a,b} \sum_{j=0}^r d_{r,j}(a,b) \zeta(r+1+s-j, c_{a,b}).$$

The $j = r$ terms have Laurent expansion $\zeta(1+s, c) = s^{-1} - \psi(c) + O(s)$. Their combined residue cancels that of the second term of $I(s)$. The finite part of the two gamma factors contributes $pq[1/r - \gamma - \log q]$ after factoring out $(r-1)!/(p^{r+1}q^2)$. Finally $F^{(r)}(p/q) = (-1)^{r+1}q^{r+1}I(0)$. All rearrangements took place in an absolutely convergent half-plane; holomorphy of the combined integral justifies taking the finite part. $\square$

Cyclotomic interpretation. For coprime p, q , the rational function

$$R_{r,p,q}(z) = \frac{\operatorname{Li}_{1-r}(z^p)}{1-z^q}$$

has poles only at p th and q th roots of unity. Its coefficient sequence has a unique finite decomposition

$$[z^N]R_{r,p,q}(z) = \sum_{\omega^p=1 \text{ or } \omega^q=1} \sum_{j=0}^r a_{\omega,j} N^j \omega^N \quad (N \geq 1).$$

The coefficients lie in $\mathbb{Q}(\mu_{pq})$, $a_{1,r} = 1/(rp^r q)$, and $a_{\omega,r} = 0$ for $\omega \neq 1$. Partial fractions therefore give the equivalent formula

$$F^{(r)}(p/q) = (-1)^{r+1}(r-1)!(q/p)^r \left(\frac{1}{r} + \log p \right) + (-1)^{r+1} r! q^{r+1} \sum_{\omega} \sum_{j=0}^{r-1} a_{\omega,j} \operatorname{Li}_{r+1-j}(\omega).$$

This proves the already known closure statement and gives all its coefficients. The special $a_{1,r}\zeta(1+s)$ term is the only pole: its finite-part contribution is exactly the displayed elementary term.

For example,

$$\begin{aligned} F''(1) &= -\frac{1}{2} - \zeta(2), & F'''(1) &= \frac{2}{3} + 3\zeta(2) + \zeta(3), \\ F''(1/2) &= -2 - 8\zeta(2) - \frac{7}{2}\zeta(3), & F'''(1/2) &= \frac{16}{3} + 48\zeta(2) + 32\zeta(3). \end{aligned}$$

At $p = q = 1$ the complete formula reduces to

$$F^{(r)}(1) = (-1)^{r+1}(r-1)! \left[\frac{1}{r} + \sum_{j=1}^{r-1} \binom{r}{j} B_{r-j}(1) \zeta(r+1-j) \right].$$

For high r this exact expression can suffer substantial floating-point cancellation; exact coefficients do not make low-precision evaluation reliable.

3.2 The complete algebraic large-order expansion

Define

$$E_r(x) = D_r(x) - \zeta(2) - \frac{x}{r}(\log x - H_{r-1}).$$

The following result concerns $r \rightarrow \infty$ with x fixed, not the large- x Bernoulli expansion of F .

Theorem 3.2 (Large derivative order with effective remainder). *For $x > 0$, integers $r \geq 2$, real $1 < A < r$, and $\pi/4 < \theta < \pi/2$,*

$$|E_r(x)| \leq \frac{2x\zeta(A)\zeta(A+1)}{\pi(2\theta - \pi/2)\cos\theta} \left(\frac{x}{2\pi\cos^2\theta} \right)^A \frac{\Gamma(r-A)\Gamma(A)\Gamma(A+1)}{\Gamma(r+1)}.$$

Consequently, uniformly for x in each compact subinterval of $(0, \infty)$,

$$E_r(x) = O\left(r^{-1/2} \exp[-2\sqrt{\pi r/x}]\right).$$

In particular, the exact harmonic-number expression $\zeta(2) + (x/r)(\log x - H_{r-1})$ contains every algebraic order in $1/r$.

Proof. Binet's digamma integral and Mellin inversion give, for $1 < c < 2$,

$$F(x) + \frac{\zeta(2)}{2x} = -\frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{\pi\Gamma(s)\zeta(s)\zeta(s+1)}{(2\pi x)^s \sin(\pi s/2)} ds.$$

One derivation is to Mellin-transform $\psi(y) - \log y + 1/(2y)$ in $1 < \Re s < 2$ using $\int_0^\infty v^{s-1}/(1+v^2) dv = \pi/[2\sin(\pi s/2)]$; the sum over n supplies $\zeta(s+1)$. Absolute convergence of Binet's integral

and exponential decay on vertical lines justify inversion and fixed-order differentiation. After r differentiations,

$$D_r(x) = \frac{\zeta(2)}{2} + \frac{1}{2\pi i} \int_{(c)} \frac{\pi \Gamma(r+s) \zeta(s) \zeta(s+1) x^{1-s}}{\Gamma(r+1) (2\pi)^s \sin(\pi s/2)} ds.$$

Move the contour to $\Re s = -A$. The only crossed poles are $s = 1$ and $s = 0$. At the negative even integers the zero of $\zeta(s)$ cancels the sine denominator; gamma poles lie to the left since $A < r$. The residue at 1 is $\zeta(2)/2$. Using $\zeta'(0) = -\frac{1}{2} \log(2\pi)$ and $\psi(r) = H_{r-1} - \gamma$, the residue at 0 is $x(\log x - H_{r-1})/r$. Thus the integral on the new line equals $E_r(x)$.

Apply the zeta functional equation twice to its kernel:

$$\frac{\pi \zeta(s) \zeta(s+1)}{(2\pi)^s \sin(\pi s/2)} = 2(2\pi)^s \cos(\pi s/2) \Gamma(1-s) \Gamma(-s) \zeta(1-s) \zeta(-s).$$

On $s = -A + it$, use $|\zeta(1-s) \zeta(-s)| \leq \zeta(A+1) \zeta(A)$ and $|\Gamma(r-A+it)| \leq \Gamma(r-A)$. Rotating the Euler gamma integral through angle $\theta \operatorname{sgn}(t)$ gives the elementary inequality

$$|\Gamma(a+it)| \leq \Gamma(a) (\cos \theta)^{-a} e^{-\theta|t|} \quad (a > 0, 0 < \theta < \pi/2).$$

Use it for the other two gamma factors, with parameters A and $A+1$, and use $|\cos(\pi s/2)| \leq e^{\pi|t|/2}$. The remaining integral is $\int_{\mathbb{R}} e^{-(2\theta-\pi/2)|t|} dt = 2/(\theta - \pi/2)$, yielding the stated effective bound. The same estimates justify the horizontal-contour limits.

For the stretched-exponential estimate take an integer $m = \sqrt{\pi r/x} + O(1)$ and $\theta = \pi/4 + 1/(2m)$. For r sufficiently large, $1 < m < r$ and the bound applies. Uniformly on compact positive x -intervals,

$$\frac{\Gamma(r-m)}{\Gamma(r+1)} = r^{-m-1} \exp \left[\frac{m(m+1)}{2r} + O(r^{-1/2}) \right], \quad \Gamma(m) \Gamma(m+1) = 2\pi m^{2m} e^{-2m} (1 + O(m^{-1})).$$

Also $(2\pi \cos^2 \theta)^{-m} = \pi^{-m} e^{1+O(m^{-1})}$, and the prefactor is $O(xm)$. The power and exponential factors reduce to $O(r^{-1/2} e^{-2\sqrt{\pi r/x}})$, as required. $\square$

An elementary factorial form. At $\theta = \pi/3$ and integer $2 \leq m < r$ the estimate becomes

$$|E_r(x)| \leq \frac{24x}{\pi^2} \zeta(m) \zeta(m+1) \left(\frac{2x}{\pi} \right)^m \frac{(r-m-1)!(m-1)!m!}{r!}.$$

It is useful as a directly computable error certificate even without asymptotic optimization.

Positivity and a convergent Taylor certificate. The derivative integral has a positive kernel with $1/2 < 1/(1 - e^{-t}) - 1/t < 1$, so

$$\frac{\zeta(2)}{2} < D_r(x) < \zeta(2).$$

Therefore the Taylor expansion of $F(x+h)$ about $x > 0$ has, for $\rho = |h|/x < 1$, the explicit remainder bound

$$\left| F(x+h) - \sum_{j=0}^N \frac{F^{(j)}(x)}{j!} h^j \right| \leq \frac{\zeta(2)}{x} \frac{\rho^{N+1}}{1-\rho}.$$

Its radius is exactly x , since $D_r(x) \rightarrow \zeta(2)$.

3.3 Exact arithmetic sectors and the leading oscillation

The divisor-weighted eta/Binet representation in Radchenko–Zagier [5, Section 7.3] is classical context for this calculation. The following identity is a differentiated and reciprocity-transformed form useful for large derivative order. The quantitative sum estimates are needed in addition to a one-sector expansion.

Write $\sigma_{-1}(n) = \sum_{d|n} d^{-1}$. Let U denote Tricomi’s confluent hypergeometric function with its principal branch. Its integral representation is DLMF 13.4.4 [8]. The large-first-parameter asymptotics used below are classical (DLMF 13.8.11–16; Temme [7]); a direct saddle calculation is included to fix the branch and the first correction in this application. The arithmetic sum over all sectors and its aggregate error bound must be justified in addition to that classical one-sector asymptotic.

Theorem 3.3 (Exact divisor-weighted expansion). *For every integer $r \geq 2$ and $x > 0$,*

$$\begin{aligned} E_r(x) &= 2x\Gamma(r) \sum_{n \geq 1} \sigma_{-1}(n) \operatorname{Re} U\left(r, 0, \frac{2\pi in}{x}\right) \\ &= 2x \sum_{n \geq 1} \sigma_{-1}(n) \int_0^\infty \frac{u^{r-1}}{(1+u)^{r+1}} \cos\left(\frac{2\pi nu}{x}\right) du. \end{aligned}$$

The series of the integrals is absolutely convergent. The assertion does not require absolute convergence after moving the infinite sum inside the oscillatory u -integral.

More precisely, for $1 < A < r$ and $\pi/4 < \theta < \pi/2$, the contribution of an arbitrary subset $\mathcal{N} \subset \mathbb{N}$ is bounded by the previous effective estimate with $\zeta(A)\zeta(A+1)$ replaced by $\sum_{n \in \mathcal{N}} \sigma_{-1}(n)n^{-A}$.

Proof. Set $w = -s$ in the post-functional-equation contour representation:

$$E_r(x) = \frac{2x}{2\pi i \Gamma(r+1)} \int_{(A)} \Gamma(r-w)\Gamma(w)\Gamma(w+1)\zeta(w)\zeta(w+1) \cos(\pi w/2) \left(\frac{x}{2\pi}\right)^w dw.$$

The Dirichlet series $\zeta(w)\zeta(w+1) = \sum_{n \geq 1} \sigma_{-1}(n)n^{-w}$ converges absolutely on this line. The effective gamma bound proves absolute convergence of the resulting series of contour integrals.

For $\Re z > 0$ put

$$M_r(z) = \int_0^\infty e^{-zu} \frac{u^{r-1}}{(1+u)^{r+1}} du = \Gamma(r)U(r, 0, z).$$

For $0 < \Re w < r$, taking its Mellin transform in z gives

$$\int_0^\infty z^{w-1} M_r(z) dz = \frac{\Gamma(w)\Gamma(r-w)\Gamma(w+1)}{\Gamma(r+1)}.$$

Mellin inversion is valid in that strip. Both the defining integral and the Barnes integral extend continuously to $z = \pm i\tau$, $\tau > 0$: the first by dominated convergence, since its rational weight is integrable, and the second by its exponential vertical decay. Averaging the two boundary values replaces z^{-w} by $\tau^{-w} \cos(\pi w/2)$. Substitution proves the formula and the termwise bounds. $\square$

A finite-sum tail bound. For an integer cutoff $N \geq 1$ and $A > 1$,

$$\sum_{n > N} \sigma_{-1}(n)n^{-A} \leq N^{1-A} \left[\frac{1 + \log N}{A-1} + \frac{1}{(A-1)^2} \right],$$

because $\sigma_{-1}(n) \leq H_n \leq 1 + \log n$ and $(1 + \log t)t^{-A}$ is decreasing for $t \geq 1$. Thus the exact series comes with a computable truncation bound.

An alternative bound avoids the auxiliary contour parameters. Rotate the defining u -integral for $M_r(i\tau)$ to the negative imaginary ray. There are no poles in that quadrant; the small and large arcs vanish. Then

$$|M_r(i\tau)| \leq \int_0^\infty e^{-\tau y} \frac{y^{r-1}}{|1 - iy|^{r+1}} dy \leq \Gamma(r)\tau^{-r}.$$

It follows that the arithmetic tail after N terms is at most

$$2x\Gamma(r) \left(\frac{x}{2\pi}\right)^r N^{1-r} \left(\frac{1 + \log N}{r-1} + \frac{1}{(r-1)^2}\right), \quad r \geq 2. \quad (3.1)$$

This is often the simplest finite-sum certificate; the earlier gamma-factor bound is more useful for the sharp large- r exponent.

Theorem 3.4 (Two-term oscillatory asymptotic). *For fixed $x > 0$ define*

$$\begin{aligned} \mathcal{A}_r(x) &= 2^{5/4}\pi^{3/4}x^{3/4}r^{-3/4}e^{-2\sqrt{\pi r/x}}, \\ \Phi_r(x) &= 2\sqrt{\pi r/x} - \frac{\pi}{x} - \frac{\pi}{8}, \\ \kappa(x) &= \frac{3}{16}\sqrt{\frac{x}{2\pi}} + \frac{1}{12}\left(\frac{2\pi}{x}\right)^{3/2}. \end{aligned}$$

Then, uniformly for x in compact subintervals of $(0, \infty)$,

$$E_r(x) = \mathcal{A}_r(x) \left[\cos \Phi_r(x) + \frac{\kappa(x)}{\sqrt{r}} \cos \left(\Phi_r(x) + \frac{\pi}{4} \right) + O(r^{-1}) \right].$$

This is an additive oscillatory expansion; a relative-error claim after division by $\cos \Phi_r(x)$ would be invalid near its zeros.

Proof. We first prove, uniformly for $z = i\tau$ with τ in any compact positive interval,

$$M_r(z) = \sqrt{\pi} z^{1/4} r^{-3/4} e^{z/2 - 2\sqrt{rz}} \left[1 + \frac{1}{\sqrt{r}} \left(\frac{3}{16\sqrt{z}} - \frac{z^{3/2}}{12} \right) + O(r^{-1}) \right]. \quad (3.2)$$

All powers use their principal values.

Rotate the u -contour in the integral defining $M_r(i\tau)$ to argument $-\pi/4$. There are no poles in that quadrant. On the large arc the rational factor is $O(R^{-2})$, while $|e^{-i\tau u}| \leq 1$; a small arc of radius ϵ contributes $O(\epsilon^r)$. Hence the rotation is justified directly. Put

$$u = \sqrt{r/z} v, \quad \lambda = \sqrt{rz}, \quad \epsilon = \sqrt{z/r}.$$

This gives the exact representation

$$M_r(z) = \sqrt{z/r} \int_0^\infty v^{-2} \exp[-\lambda v - (r+1)\log(1 + \epsilon/v)] dv.$$

On any fixed compact positive v -interval,

$$\begin{aligned} -\lambda v - (r+1)\log(1 + \epsilon/v) &= -\lambda(v + v^{-1}) + \frac{z}{2v^2} \\ &\quad + r^{-1/2} \left(-\frac{\sqrt{z}}{v} - \frac{z^{3/2}}{3v^3} \right) + O(r^{-1}), \end{aligned}$$

with the same estimates after any fixed number of v -derivatives.

For completeness, localization at $v = 1$ is uniform. Write $a = \sqrt{\tau/(2r)}$, so $\epsilon = a(1 + i)$ and $\Re \lambda = ra$. On a compact v -interval the real part of the exponent is $-ra(v + v^{-1}) + O(1)$, and $v + v^{-1}$ has its unique minimum 2 at 1. For the small tail $0 < v < \delta < 1/2$, use $|1 + \epsilon/v| \geq 1 + a/v$ and substitute $y = a/v$:

$$\int_0^\delta v^{-2} |1 + \epsilon/v|^{-r-1} dv \leq \frac{1}{ar} (1 + a/\delta)^{-r} = O\left(r^{-1/2} e^{-ra/\delta + O(1)}\right).$$

This is exponentially smaller than e^{-2ra} . For $v > M > 2$, the factor $e^{-\lambda v}$ gives the analogous bound $O(e^{-Mra})$. The remaining compact interval outside any fixed neighborhood of 1 also has a strict exponential gap.

Apply complex Laplace expansion at $v = 1$ to $\phi(v) = v + v^{-1}$ and $g(v) = v^{-2} e^{z/(2v^2)}$. It is legitimate since $\arg \lambda = \pi/4$ lies strictly inside the right half-plane. The Gaussian integral along the real v -axis gives $e^{-2\lambda} \sqrt{\pi/\lambda}$. The coefficient of $1/\lambda$, divided by $g(1)$, is

$$\frac{g''(1)}{4g(1)} + \frac{3g'(1)}{4g(1)} + \frac{3}{16} = z + \frac{z^2}{4} + \frac{3}{16}.$$

Adding the explicit $r^{-1/2}$ correction in the exponent at $v = 1$ gives

$$-\sqrt{z} - \frac{z^{3/2}}{3} + \frac{1}{\sqrt{z}} \left(z + \frac{z^2}{4} + \frac{3}{16} \right) = \frac{3}{16\sqrt{z}} - \frac{z^{3/2}}{12}.$$

This proves (3.2); further terms follow by the same finite Taylor and Gaussian-moment calculation, with standard local remainder estimates. As an independent classical check, in DLMF 13.8.11 with $b = 0$ one has $p_0(z) = 1$ and $q_0(z) = -z/12$; the standard large-argument expansions of K_1 and K_0 give exactly the same first correction.

At $z = 2\pi i/x$, taking $2x$ times the real part of (3.2) gives the stated main term and correction. Indeed $z^{1/4}$ has argument $\pi/8$, and the correction equals $\kappa(x)e^{-\pi i/4}$.

It remains to control all the other arithmetic sectors uniformly. For $A \geq 2$,

$$\zeta(A)\zeta(A+1) - 1 = \sum_{n \geq 2} \sigma_{-1}(n)n^{-A} \leq 4(\zeta(2)\zeta(3) - 1)2^{-A}.$$

Use the effective sector bound with $A = \sqrt{2\pi r/x} + O(1)$ an integer and $\theta = \pi/4 + 1/(2A)$. Exactly the previous optimization yields

$$\sum_{n \geq 2} 2x\sigma_{-1}(n) \operatorname{Re} M_r(2\pi in/x) = O\left(r^{-1/2} e^{-2\sqrt{2\pi r/x}}\right).$$

This is absorbed into $\mathcal{A}_r(x)O(r^{-1})$, uniformly on compact positive x -intervals. $\square$

Corollary 3.5 (Sharp exponential rate and infinitely many signs). *For each fixed $x > 0$ the real sequence $E_r(x)$ takes positive and negative values infinitely often. More precisely,*

$$\limsup_{r \rightarrow \infty} \frac{E_r(x)}{\mathcal{A}_r(x)} = 1, \quad \liminf_{r \rightarrow \infty} \frac{E_r(x)}{\mathcal{A}_r(x)} = -1,$$

and

$$\limsup_{r \rightarrow \infty} \frac{\log |E_r(x)|}{\sqrt{r}} = -2\sqrt{\pi/x}.$$

Here $\log 0 = -\infty$ if an exact zero occurs.

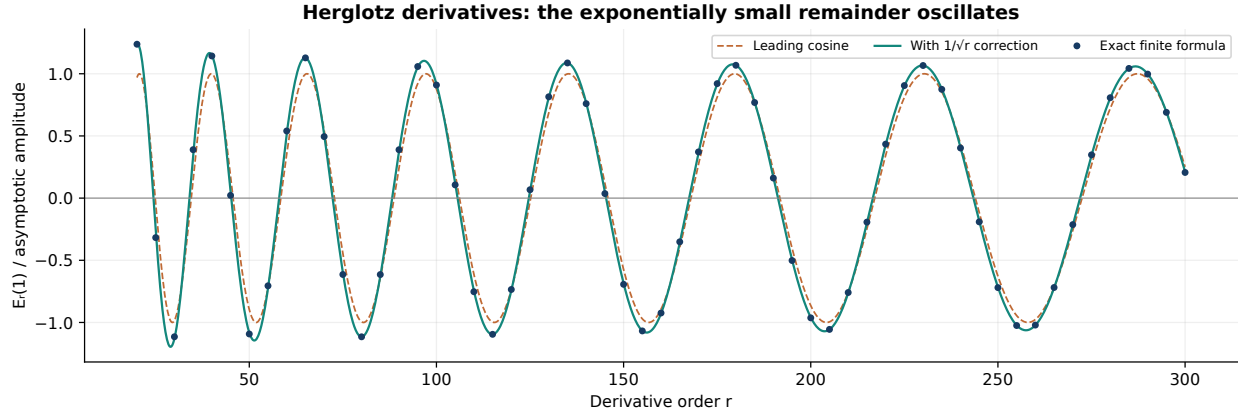


Figure 2: The Herglotz remainder divided by its asymptotic amplitude at $x = 1$. The leading cosine and its first correction track the exact finite-formula values. The asymptotic is additive, so no division by a small cosine is used.

Proof. The phase $\Phi_r(x)$ increases to infinity, with successive differences tending to zero. Consequently it has subsequences approaching any fixed phase modulo 2π . Apply the additive expansion and choose phases 0 and π . The same subsequences establish the lower bound for the limsup of the logarithmic rate; the asymptotic envelope gives the upper bound. $\square$

Further research specific to this section. Compute and certify several further half-integer corrections in (3.2); build an interval implementation of the exact sector sum; locate the sign-transition indices beyond the leading $\sqrt{r}$ phase law; and obtain a uniform transition theory when x also varies with r . The fixed- x theorem above does not justify such a double-scaling limit automatically.

4 Cyclotomic double relations: exact identities and formal obstructions

Set

$$g_{a,b} = \operatorname{Im} \operatorname{Li}_{a,b}(i, 1), \quad A_{a,b} = \operatorname{Im} \operatorname{Li}_{a,b}(i, i), \quad K_{a,b} = \operatorname{Im} \operatorname{Li}_{a,b}(i, -1).$$

The manuscript's mixed harmonic sums are

$$S_p = \sum_{n \geq 0} \frac{(-1)^n H_n}{(2n+1)^p}, \quad p \geq 1.$$

For $p > 1$ this converges absolutely; at $p = 1$ it converges conditionally. The all-odd-index evaluation of S_p has already been proved in the repository's continuations. Our questions concern even indices and what exact relations can establish about them.

4.1 Two exact weight-five reductions

Proposition 4.1 (Two coordinates suffice for the weight-five g -family). *The four values of weight five satisfy*

$$g_{2,3} + 3g_{3,2} + 6g_{4,1} = -\frac{3}{32}G\zeta(3) - \frac{\pi^5}{1536}, \quad (4.1)$$

$$g_{1,4} + g_{2,3} + g_{3,2} + 2g_{4,1} = -\frac{1}{2}\beta(4)\log 2 - \frac{7\pi^5}{46080}. \quad (4.2)$$

Thus they lie in the span of any two suitably chosen coordinates, for example $g_{4,1}, g_{2,3}$, together with the displayed products of single values. This is a spanning statement, not an independence statement.

Proof. Shuffle the two iterated-integral words for $\text{Li}_2(i) \text{Li}_3(i)$. Their imaginary part is $g_{2,3} + 3g_{3,2} + 6g_{4,1}$. Since

$$\text{Li}_2(i) = -\frac{\zeta(2)}{8} + iG, \quad \text{Li}_3(i) = -\frac{3\zeta(3)}{32} + i\frac{\pi^3}{32},$$

its product gives (4.1). Likewise, the shuffle of $\text{Li}_1(i) \text{Li}_4(i)$ has imaginary part $g_{1,4} + g_{2,3} + g_{3,2} + 2g_{4,1}$. Use

$$\text{Li}_1(i) = -\frac{1}{2}\log 2 + i\frac{\pi}{4}, \quad \text{Li}_4(i) = -\frac{7}{128}\zeta(4) + i\beta(4).$$

The two rows have rank two already in their g -coefficients. $\square$

The manuscript's displayed sporadic row

$$576g_{4,1} + 288g_{3,2} + 736g_{2,3} + 960g_{1,4} = 21G\zeta(3) - 480\beta(4)\log 2$$

is -224 times (4.1) plus 960 times (4.2). It therefore has a direct proof from these two elementary shuffles. The manuscript's *at most three* statement can be strengthened to *at most two*. Existing continuations already address proofs of the sporadic relation; the point here is the sharper two-row organization.

4.2 An all-exponent mixed endpoint identity

Theorem 4.2 (Gaussian mixed endpoint). *For every integer $p \geq 1$,*

$$K_{p,1} + (-1)^{p-1}A_{1,p} = \beta(p+1) - 2\log 2 \beta(p) + \sum_{j=2}^p (-1)^j \eta(j) \beta(p+1-j). \quad (4.3)$$

Moreover,

$$S_p = g_{p,1} + K_{p,1}. \quad (4.4)$$

Proof. All the boundary values used here converge. First-index-one values with first argument i can be obtained by Dirichlet's test or radial Abel limits. The shuffle and stuffle identities may therefore be justified by convergent iterated integrals and Abel limiting.

Fix $w = p + 1$, and define homogeneous generating polynomials of degree $w - 2$:

$$\mathcal{A}(X, Y) = \sum_{a+b=w} A_{a,b} X^{a-1} Y^{b-1}, \quad \mathcal{K}(X, Y) = \sum_{a+b=w} K_{a,b} X^{a-1} Y^{b-1}.$$

Let $\mathcal{V}$ denote the same polynomial for $\text{Im Li}_{a,b}(-1, i)$, and put

$$P(X, Y) = \sum_{a+b=w} \text{Im}[\text{Li}_a(-1) \text{Li}_b(i)] X^{a-1} Y^{b-1}, \quad H(X, Y) = \sum_{a+b=w} X^{a-1} Y^{b-1}.$$

Stuffle and shuffle respectively give

$$\begin{aligned} \mathcal{V}(X, Y) + \mathcal{H}(Y, X) &= P(X, Y) + \beta(w)H(X, Y), \\ \mathcal{A}(X + Y, X) - \mathcal{V}(X + Y, Y) &= P(X, Y). \end{aligned}$$

The collision term in the first equation is $-\text{Im Li}_w(-i) = \beta(w)$. Eliminating $\mathcal{V}$ yields

$$\mathcal{H}(U, V) = P(V - U, U) + P(V, U) + \beta(w)H(V, U) - \mathcal{A}(V, V - U).$$

Set $V = 0$ and compare coefficients of U^{w-2} . The evaluation $\text{Li}_j(-1) = -\eta(j)$ gives (4.3), including the two occurrences of $-\log 2 \beta(p)$.

For (4.4), taking imaginary parts of the outer index in $\text{Li}_{p,1}(i, 1) + \text{Li}_{p,1}(i, -1)$ retains indices $2n + 1$, with sign $(-1)^n$. Their inner sum is

$$\sum_{m=1}^{2n} \frac{1 + (-1)^m}{m} = H_n.$$

Absolute convergence proves the result directly for $p > 1$. An Abel limit gives the endpoint $p = 1$. $\square$

At $p = 4$ this specializes to

$$K_{4,1} = A_{1,4} + \frac{191\pi^5}{23040} - \frac{3}{4}G\zeta(3) - 2\beta(4)\log 2. \quad (4.5)$$

4.3 Inverse-argument mixed values and a source correction

A useful regularization identity also removes a specific proposed mixed survivor from the manuscript. Inverse-argument double reductions are part of the existing repository programme; the following short derivation makes the correction explicit.

Proposition 4.3 (Inverse-argument endpoint). *For $p \geq 1$ and $z = i$ or $e^{2\pi i/3}$,*

$$\begin{aligned} \text{Im Li}_{p,1}(z, z^{-1}) &= \sum_{a=1}^p \text{Im Li}_{a,p+1-a}(z, 1) + p \text{Im Li}_{p+1}(z) \\ &\quad - \sum_{j=2}^p \zeta(j) \text{Im Li}_{p+1-j}(z). \end{aligned} \quad (4.6)$$

Proof. Give $\text{Li}_1(1)$ its usual real regularization parameter T . The shuffle expansion of $T \text{Li}_p(z)$ is

$$\text{Li}_{p,1}(z, z^{-1}) + \sum_{a=1}^p \text{Li}_{a,p+1-a}(1, z),$$

and its shuffle expansion is

$$\text{Li}_{1,p}(1, z) + \text{Li}_{p,1}(z, 1) + \text{Li}_{p+1}(z).$$

The divergent leading term cancels on subtraction. For $2 \leq a \leq p$ use the convergent stuffle

$$\text{Li}_{a,p+1-a}(1, z) = \zeta(a) \text{Li}_{p+1-a}(z) - \text{Li}_{p+1-a,a}(z, 1) - \text{Li}_{p+1}(z).$$

Taking imaginary parts gives (4.6). For $p = 1$ there is only one divergent factor because $z \neq 1$; the same cancellation proves the formula. $\square$

Substitution of the already proved Gaussian weight-six g -values gives the particularly short identity

$$\boxed{\operatorname{Im} \operatorname{Li}_{5,1}(i, -i) = 3\beta(6) - \frac{\pi^2}{6}\beta(4) - \frac{\pi^4}{90}G - \frac{5\pi^5}{3072}\log 2.} \quad (4.7)$$

Thus this imaginary weight-six value is in the depth-one product basket. The candidate-survivor wording in Chapter 4 should be removed.

For $\omega = e^{2\pi i/3}$, define $C_s = \operatorname{Im} \operatorname{Li}_s(\omega)$; this is the sine-part convention for every integer s . Then

$$\begin{aligned} \operatorname{Im} \operatorname{Li}_{5,1}(\omega, \omega^2) &= \sum_{a=1}^5 \operatorname{Im} \operatorname{Li}_{a,6-a}(\omega, 1) + 5C_6 - \frac{\pi^2}{6}C_4 - \frac{\pi^4}{90}C_2 \\ &\quad - \frac{2\pi^3}{81}\zeta(3) - \frac{\pi}{6}\zeta(5). \end{aligned} \quad (4.8)$$

Here $C_1 = \pi/6$ and $C_3 = 2\pi^3/81$. It follows that the corresponding level-three imaginary value contributes no new direction modulo the stated g -family and single products. This conclusion says nothing about the manuscript's real weight-five candidate.

4.4 A precisely defined formal presentation

Fix a level $L = 3$ or 4 and a weight $w \geq 2$. Let $\omega = e^{2\pi i/L}$. Introduce rational formal symbols

$$D_a(r, s) \quad (1 \leq a < w, \quad r, s \in \mathbb{Z}/L\mathbb{Z})$$

for imaginary parts of $\operatorname{Li}_{a,w-a}(\omega^r, \omega^s)$. Impose conjugation $D_a(-r, -s) = -D_a(r, s)$. Symbols fixed by this involution vanish. Put all single values and imaginary parts of products of two single values in the background, and also set $D_a(1, 0) = 0$. The latter is precisely quotienting by the chosen g -family.

The only further rows imposed are, for $p + q = w$,

$$D_p(r, s) + D_q(s, r) = 0, \quad (4.9)$$

$$\sum_{j=0}^{p-1} \binom{q+j-1}{j} D_{q+j}(s, r-s) + \sum_{j=0}^{q-1} \binom{p+j-1}{j} D_{p+j}(r, s-r) = 0. \quad (4.10)$$

They are the stuffle and shuffle rows of two single values, with their one-leading-one regularizations. A regularized product involving the real parameter T belongs to the background. At $w = 2$, the only two-divergent-factor correction is real and disappears on taking imaginary parts. Call the resulting quotient $V_{L,w}$.

This is *not* the full cyclotomic double-shuffle algebra. Products involving a depth-two factor, relations lifted through higher depths, additional functional equations, and motivic period assertions have not been imposed. In particular, Zhao's standard relations [9] are substantially broader than this presentation. The theorem below computes $V_{L,w}$, not a dimension of actual numbers.

For a scalar assignment to the symbols, introduce

$$\mathcal{F}^{r,s}(X, Y) = \sum_{a=1}^{w-1} D_a(r, s) X^{a-1} Y^{w-a-1}.$$

The conditions that the assignment annihilate all rows are exactly

$$\mathcal{F}^{r,s}(X, Y) + \mathcal{F}^{s,r}(Y, X) = 0, \quad (4.11)$$

$$\mathcal{F}^{s,r-s}(X+Y, X) + \mathcal{F}^{r,s-r}(X+Y, Y) = 0. \quad (4.12)$$

These follow by comparing binomial coefficients after substitution. They describe the dual of $V_{L,w}$, so their solution dimension is the required quotient dimension.

4.5 All-weight dimension theorems

Theorem 4.4 (Level four). *The formal quotient satisfies*

$$\dim_{\mathbb{Q}} V_{4,w} = \begin{cases} 0, & w \text{ even}, \\ (w-1)/2, & w \text{ odd}. \end{cases} \quad \sum_{w \geq 2} \dim V_{4,w} t^w = \frac{t^3}{(1-t^2)^2}.$$

At odd weight, its dual is parameterized by arbitrary antisymmetric homogeneous polynomials of degree $w-2$.

Proof. Up to conjugation, the six color pairs are $(0,1), (1,0), (1,1), (1,2), (1,3), (2,1)$. Write their polynomials as U, G, A, K, M, V , respectively, and let $n = w-2$. The background condition gives $G=0$; stuffle gives

$$U=0, \quad A(Y,X) = -A(X,Y), \quad V(X,Y) = -K(Y,X), \quad M(Y,X) = M(X,Y).$$

The shuffle with colors $(1,0)$ forces $M=0$. The shuffle with colors $(2,1)$ gives

$$A(X+Y,X) + K(Y,X+Y) = 0,$$

hence $K(X,Y) = -A(Y,Y-X)$. The shuffle with colors $(1,3)$ is $K(X,Y) = K(X,X-Y)$. After substitution and antisymmetry this condition is $A = (-1)^{n+1}A$.

For even n all polynomials therefore vanish. For odd n , any antisymmetric A , with K, V as above and $U = G = M = 0$, satisfies the rows. To check exhaustiveness, the unordered color classes up to conjugation are

$$(0,0), (0,1), (0,2), (1,1), (1,2), (1,3), (2,2).$$

The real classes vanish, and the remaining classes are precisely the conditions just used, together with their transposes. Thus there are no unexamined rows. A basis for A at odd w is

$$X^{a-1}Y^{w-a-1} - X^{w-a-1}Y^{a-1}, \quad 1 \leq a \leq (w-1)/2.$$

Counting these basis vectors and summing over odd weights proves the formulas. $\square$

Theorem 4.5 (Level three). *The formal quotient satisfies*

$$\dim_{\mathbb{Q}} V_{3,w} = \begin{cases} 0, & w \text{ even}, \\ \lfloor (w+1)/6 \rfloor, & w \text{ odd}, \end{cases} \quad \sum_{w \geq 2} \dim V_{3,w} t^w = \frac{t^5}{(1-t^2)(1-t^6)}.$$

Writing $\Delta = XY(X-Y)$, $Q = X^2 - XY + Y^2$, the dual is parameterized by the homogeneous polynomials

$$A(X,Y) = \Delta P(Q, \Delta^2)$$

of degree $w-2$.

Proof. Up to conjugation the four pairs are $(0,1), (1,0), (1,1), (1,2)$. As before the first two vanish and the shuffle with a zeta factor kills the inverse-argument polynomial. The remaining polynomial A satisfies exactly

$$A(Y,X) = -A(X,Y), \quad A(X,X-Y) = A(X,Y).$$

Antisymmetry implies divisibility by $X - Y$; the second symmetry then forces divisibility by Y , and antisymmetry forces divisibility by X . Hence $A = \Delta B$. The polynomial Δ has the same two transformation signs as A , so B is invariant under swapping X, Y and under $(X, Y) \mapsto (X, X - Y)$.

On the plane $a + b + c = 0$, with $(a, b, c) = (X, -Y, Y - X)$, the second transformation swaps b, c ; the first is global negation followed by the transposition of a, b . The generated group contains global negation, obtained by cubing the product, and all permutations. The elementary symmetric-polynomial theorem therefore identifies its invariant ring with

$$\mathbb{Q}[e_2, e_3^2] = \mathbb{Q}[Q, \Delta^2].$$

Conversely every $\Delta P(Q, \Delta^2)$ has the required symmetries, and direct substitution verifies all the original color rows. Counting monomials of degrees $3 + 2a + 6b = w - 2$ gives the result. Explicitly, at odd $w \geq 5$ a basis is

$$\Delta^{2b+1} Q^{(w-5)/2-3b}, \quad 0 \leq b \leq \lfloor (w-5)/6 \rfloor.$$

□

For comparison, the exact dimensions at the first few odd weights are

w	3	5	7	9	11	13	15
$\dim V_{3,w}$	0	1	1	1	2	2	2
$\dim V_{4,w}$	1	2	3	4	5	6	7

and all even-weight dimensions are zero. The exact matrix script checks both formulas through weight 16, including that the displayed polynomial bases annihilate every raw row and have the stated rank.

4.6 A separating functional for the S_4 conjecture

At every odd weight $w \geq 3$, choose $A(X, Y) = (Y - X)^{w-2}$ in the level-four theorem. Then $K(X, Y) = X^{w-2}$, $V(X, Y) = -Y^{w-2}$. This gives a rational linear functional ℓ that annihilates every defining row and every background value:

$$\begin{aligned} \ell(D_a(1, 1)) &= (-1)^{a+1} \binom{w-2}{a-1}, \\ \ell(D_{w-1}(1, 2)) &= 1, \quad \ell(D_1(2, 1)) = -1. \end{aligned} \tag{4.13}$$

All other entries vanish, apart from conjugates with opposite signs.

At weight five its nonzero entries are particularly transparent:

symbol	$A_{1,4}$	$A_{2,3}$	$A_{3,2}$	$A_{4,1}$	$\text{Im Li}_{1,4}(-1, i)$	$K_{4,1}$
ℓ	1	-3	3	-1	-1	1

The polynomial proof verifies all weights; the delivered certificate also checks this six-entry vector against every exact weight-five row.

Conjecture 4.6 (The manuscript's S_4 identity). The following identity remains unproved in this continuation:

$$S_4 = \frac{4g_{4,1} - 3g_{3,2} - 9g_{2,3}}{7} + \frac{\pi^5}{224} - \frac{27}{224} G\zeta(3) - 2\beta(4) \log 2. \tag{4.14}$$

The exact identity (4.5) makes this equivalent to

$$161280A_{1,4} + 69120(g_{4,1} + g_{3,2} + 3g_{2,3}) + 617\pi^5 - 101520G\zeta(3) = 0. \quad (4.15)$$

The functional ℓ sends its left side to 161280. Consequently this relation does not follow from (4.9)–(4.10) and the stated background alone. This is an exact obstruction to a specific proof strategy. Additional true analytic relations may still prove it.

Using (4.1), an equivalent version with only two g -coordinates is

$$S_4 = \frac{10g_{4,1} - 8g_{2,3}}{7} + \frac{7\pi^5}{1536} - \frac{3}{28}G\zeta(3) - 2\beta(4)\log 2. \quad (4.16)$$

All three displayed candidates are equivalent by proved identities; none is silently promoted to a theorem.

4.7 A new weight-seven candidate

Conjecture 4.7 (A reduction for S_6). With the same definitions,

$$\begin{aligned} S_6 = & \frac{722}{527}g_{6,1} + \frac{40}{527}g_{4,3} - \frac{128}{155}g_{2,5} + \frac{15191}{28569600}\pi^7 \\ & - \frac{3}{124}G\zeta(5) - \frac{2373}{10540}\beta(4)\zeta(3) - 2\beta(6)\log 2. \end{aligned} \quad (4.17)$$

The candidate was found in the ordered basket

$$(S_6, g_{6,1}, g_{4,3}, g_{2,5}, \pi^7, G\zeta(5), \beta(4)\zeta(3), \beta(6)\log 2).$$

Its primitive integer relation vector is

$$(485683200, -665395200, -36864000, 401080320, \\ -258247, 11750400, 109347840, 971366400).$$

A 100-digit search used coefficient bound 10^{12} and tolerance 10^{-90} . After the coefficients were frozen, reevaluation by Mellin integrals at 220 working digits gave an integer-normalized residual of approximately -2.315×10^{-213} . An independent Euler transformation of the defining alternating series, using 900 terms and 300 working digits, gave approximately 2.854×10^{-263} . The script records the full values and the exact integer vector. These are numerical evidence, not a proof or an interval certificate.

For the second method, the defining series are

$$g_{a,b} = \sum_{n \geq 0} \frac{(-1)^n H_{2n}^{(b)}}{(2n+1)^a}, \quad H_m^{(b)} = \sum_{j=1}^m j^{-b},$$

and the analogous series with H_n for S_6 . Thus the second evaluation does not reuse the discovery integral. Last transformed terms are recorded as diagnostics; their small size alone is not asserted to bound the complete tail.

The weight-seven functional (4.13) sends the integer-normalized candidate to 485683200. Its proof therefore also requires information beyond the restricted single-product presentation. A useful next step is to obtain a functional equation that proves both (4.14) and (4.17), with the nonzero functional serving as a check that the missing relation has actually been introduced.

4.8 A rigorous rational certificate for the new conjecture

The floating-point checks above can be strengthened to an exact residual enclosure without resolving the equality itself.

Theorem 4.8 (Euler remainder for harmonic numerators). *Let a, b, q be positive integers, $c_n = H_{qn}^{(b)} / (2n+1)^a$, and $L = \sum_{n \geq 0} (-1)^n c_n$. Define $\Delta c_n = c_{n+1} - c_n$ and*

$$E_N = \sum_{k=0}^{N-1} \frac{(-\Delta)^k c_0}{2^{k+1}}, \quad N \geq 1.$$

Then

$$0 \leq E_N - L \leq (N+1)2^{-N} c_1 = (N+1)2^{-N} \frac{H_q^{(b)}}{3^a}. \quad (4.18)$$

Proof. On the unit square use the positive measure

$$d\mu(t, u) = \frac{(-\log t)^{a-1} (-\log u)^{b-1}}{(a-1)!(b-1)!(1-u)} dt du.$$

Set $v = t^2$, $z = t^2 u^q$. Integral representations for reciprocal integer powers give

$$c_n = \int (v^n - z^n) d\mu, \quad \int (v - z) d\mu = c_1 < \infty.$$

The measure itself need not have finite mass. All differences below are integrable because they are bounded by a constant times $v - z$. For $k \geq 1$,

$$(-\Delta)^k c_0 = \int [(1-v)^k - (1-z)^k] d\mu \leq 0, \quad |(-\Delta)^k c_0| \leq k c_1.$$

Thus the Euler series converges absolutely. It equals L : introduce an Abel factor into the original series, sum the geometric differences under an integrable bound, and let the factor increase to 1. The ordinary series converges absolutely for $a > 1$. When $a = 1$, its magnitudes eventually decrease to zero: the harmonic increment divided by $H_{qn}^{(b)}$ is $O(1/(n \log n))$ for $b = 1$ and $O(n^{-b})$ for $b > 1$, whereas the denominator ratio is $1 + 2/(2n+1)$. Abel's theorem identifies the limits. Every omitted Euler term is nonpositive, and

$$\sum_{k=N}^{\infty} \frac{k}{2^{k+1}} = (N+1)2^{-N}.$$

This proves both the sign and the bound. □

The finite sum has the exact rational form

$$E_N = 2^{-N} \sum_{n=0}^{N-1} (-1)^n c_n \sum_{j=n+1}^N \binom{N}{j}. \quad (4.19)$$

Indeed, $\sum_{k=n}^{N-1} \binom{k}{n} 2^{-k-1} = 2^{-N} \sum_{j=n+1}^N \binom{N}{j}$. When $q = 1$ or 2 , a common denominator is $2^N \text{lcm}(1, 2, \dots, 2N)^{a+b}$. Thus all four nonclassical constants in the S_6 basket have rapid exact rational enclosures.

For $c_n = (dn+1)^{-s}$, $d = 1$ or 2 , a positive moment measure of total mass 1 instead gives $0 \leq L - E_N \leq 2^{-N}$. This encloses $\eta(s)$, $\beta(s)$, and hence $\zeta(3)$, $\zeta(5)$, $\log 2$ using rational factors. For π use Machin's identity

$$\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$$

and the alternating-series bound for each arctangent.

The delivered standard-library script `certify_s6.py` uses $N = 900$, 240 arctangent terms at $1/5$, and 80 at $1/239$. It performs every interval operation with integers and rational fractions, and rounds final decimal endpoints outward. The resulting integer-normalized residual R_6 satisfies the deliberately widened, easily readable rational enclosure

$$-5.016 \times 10^{-261} \leq R_6 \leq 3.845 \times 10^{-262}. \quad (4.20)$$

The full 280-place endpoints are recorded in the JSON certificate. The exact arithmetic gate checks, in particular,

$$\boxed{|R_6| < 10^{-260}}.$$

This is a rigorous error statement about the residual. It is still not a proof that the residual equals zero.

5 The cubic log-gamma moment: a classical evaluation and a source correction

Write

$$a = \frac{1}{2} \log(2\pi), \quad A = \gamma + \log(2\pi), \quad M_j = \int_0^1 \log^j \Gamma(x) dx.$$

The manuscript's statement that identifying a named depth-two atom for M_3 is open requires correction. Bailey, D. Borwein, and J. M. Borwein evaluated M_3 using derivatives of Tornheim–Witten zeta functions in Theorem 6, equation (61), of their 2015 paper [4]. The author-hosted preprint is dated 27 June 2012. Their formula does not establish a reduction to a prescribed finite collection of ordinary zeta values and derivatives. That more restrictive question should be kept distinct from the already established Tornheim evaluation. The argument below independently derives an equivalent formula and supplies a convenient convergence justification.

For complex parameters in an open neighbourhood of $(1, 1, 1)$, define

$$T(s, t, u) = \sum_{m, n \geq 1} \frac{1}{m^s n^t (m+n)^u}, \quad T_j = \partial_j T, \quad T_{ij} = \partial_i \partial_j T.$$

Every derivative below is evaluated at $(1, 1, 1)$ unless arguments are shown. The notation T_{12} and T_{13} is preferable to an unqualified prime, because different parameter derivatives give different constants.

Theorem 5.1 (Classical cubic evaluation, with a direct proof). *Set*

$$V = \frac{\pi^2}{48} + \frac{A^2}{12} - \frac{A\zeta'(2)}{\pi^2} + \frac{\zeta''(2)}{2\pi^2}.$$

Then

$$\boxed{M_3 = a^3 + 3aV + \frac{3\zeta(3)}{16} + \frac{3}{8\pi^2} (2A^2\zeta(3) - 2AT_3 + 2T_{13} - T_{12})}. \quad (5.1)$$

All derivatives on the right are represented by absolutely convergent double series. In particular, no analytic continuation or summation prescription is needed to define these constants.

Proof. Let $f(x) = \log \Gamma(x)$, regarded as an integrable periodic function. Since $f(x) = -\log x + O(x)$ as $x \downarrow 0$ and f is bounded near 1, we have $f \in L^p(0, 1)$ for every finite $p \geq 1$. Kummer's Fourier expansion gives

$$f(x) = a + \sum_{n \geq 1} \left(\frac{\cos(2\pi nx)}{2n} + \frac{A + \log n}{\pi n} \sin(2\pi nx) \right), \quad 0 < x < 1.$$

Equivalently, the positive Fourier coefficients are

$$h_n = \frac{1}{n} \left(\frac{1}{4} - \frac{i(A + \log n)}{2\pi} \right), \quad h_{-n} = \overline{h_n}.$$

For $0 < r < 1$, let f_r be the Poisson mean of f , so that the coefficient h_n is replaced by $r^{|n|}h_n$. The Fourier series is now absolutely convergent, and Poisson approximate-identity convergence gives $f_r \rightarrow f$ in L^3 . Consequently $\int f_r^j \rightarrow \int f^j$ for $j = 1, 2, 3$: for the cubic case this follows, for example, from

$$\|u^3 - v^3\|_1 \leq \|u - v\|_3 (\|u\|_3^2 + \|u\|_3 \|v\|_3 + \|v\|_3^2).$$

Write $g = f - a$. Orthogonality first gives

$$\begin{aligned} \int_0^1 g(x)^2 dx &= 2 \sum_{n \geq 1} |h_n|^2 \\ &= \frac{\zeta(2)}{8} + \frac{A^2 \zeta(2) - 2A\zeta'(2) + \zeta''(2)}{2\pi^2} = V. \end{aligned}$$

For the cube, a zero-sum triple of nonzero Fourier frequencies has two positive frequencies m, n and one negative frequency $-(m+n)$, or the opposite signs. Accounting for the three positions of the exceptional sign and then conjugation gives

$$\int_0^1 g_r(x)^3 dx = 6 \operatorname{Re} \sum_{m, n \geq 1} r^{2(m+n)} h_m h_n \overline{h_{m+n}}. \quad (5.2)$$

The factor is correct also when $m = n$: the ordered double sum accounts for the interchange of the two positive frequencies only when they differ.

These resonant sums converge absolutely at $r = 1$. Indeed,

$$\sum_{m=1}^{k-1} \frac{1}{m(k-m)k} = \frac{2H_{k-1}}{k^2},$$

and $|h_m h_n h_{m+n}|$ is bounded by a constant times $(1 + \log(m+n))^3 / [mn(m+n)]$. The grouped majorant is $O((1 + \log k)^4 / k^2)$, which is summable. Dominated convergence therefore permits $r \uparrow 1$ in (5.2).

Put $p = 1/4$ and $q_j = (A + \log j)/(2\pi)$. Direct multiplication yields

$$\operatorname{Re}[(p - iq_m)(p - iq_n)(p + iq_{m+n})] = p^3 + p(q_m q_{m+n} + q_n q_{m+n} - q_m q_n).$$

The numerator in the second term is, apart from $4\pi^2$,

$$A^2 + 2A \log(m+n) + \log m \log(m+n) + \log n \log(m+n) - \log m \log n.$$

By the definitions of the Tornheim derivatives, symmetry in m, n , and $T(1, 1, 1) = 2\zeta(3)$, we obtain

$$\int_0^1 g(x)^3 dx = \frac{3\zeta(3)}{16} + \frac{3}{8\pi^2} (2A^2 \zeta(3) - 2AT_3 + 2T_{13} - T_{12}).$$

Finally, $\int g = 0$ and $\int f^3 = a^3 + 3a \int g^2 + \int g^3$. □

Holomorphy and differentiation. For completeness, local holomorphy is also elementary. If each real part of s, t, u is at least $1 - \epsilon$, with $0 < \epsilon < 1/3$, grouping by $k = m + n$ and splitting at $m = k/2$ gives a locally uniform majorant of order $k^{-2+3\epsilon}$. The sum converges uniformly on compact subsets of such a neighbourhood. Standard termwise differentiation therefore applies. Alternatively, at $(1, 1, 1)$ any prescribed logarithmic numerator of degree d is dominated by $2H_{k-1}(\log k)^d/k^2$ after grouping. These are ordinary convergent derivatives; Abel means are used only to justify the Fourier multiplication.

Relations to Euler double zeta derivatives. Define $Z(s) = \zeta(s, 1) = \sum_{k \geq 2} H_{k-1} k^{-s}$, so that the prime in $Z'(2)$ differentiates its first argument. Then

$$T(1, 1, u) = 2Z(u + 1), \quad T_3 = 2Z'(2), \quad T_1 = T_2 = Z'(2) + \zeta'(3). \quad (5.3)$$

The first identity follows from $1/[m(k - m)] = (1/m + 1/(k - m))/k$. For the last identity, use $\sum_{n \geq 1} 1/[n(m + n)] = H_m/m$. Equation (5.1) thus exactly matches the published cubic formula. It also makes clear why a search involving only ordinary Riemann-zeta derivatives need not find a relation. Failure of a finite PSLQ search does not prove that such a reduction is impossible.

5.1 A positive summand form and an explicit tail bound

The combination in (5.1) can be computed without cancellation between separate Tornheim constants. For $k = m + n$, define

$$R_A(m, n) = A^2 + 2A \log k + \log m \log k + \log n \log k - \log m \log n.$$

Since $A > 0$, $\log k \geq \log m, \log n \geq 0$, we have $R_A(m, n) > 0$. Writing $p = \log m, q = \log n, r = \log k$, one also has $r(p + q) - pq \leq r^2$: for fixed r , the expression is increasing separately in $p, q \in [0, r]$. Consequently

$$0 < R_A(m, n) \leq (A + \log k)^2.$$

Therefore the diagonal partial sums

$$C_N = \frac{3}{32} \sum_{m+n \leq N} \frac{1}{mn(m+n)} + \frac{3}{8\pi^2} \sum_{m+n \leq N} \frac{R_A(m, n)}{mn(m+n)}$$

increase to the centered cubic moment $M_3 - a^3 - 3aV$. Moreover,

$$0 < (M_3 - a^3 - 3aV) - C_N \leq \sum_{k > N} \frac{2(1 + \log k)}{k^2} \left(\frac{3}{32} + \frac{3(A + \log k)^2}{8\pi^2} \right).$$

This gives a transparent computable enclosure, although its convergence is only $O(\log^3 N/N)$. An integral bound may be used once the explicit summand is decreasing; direct summation can handle the finite initial range. This positivity observation is an immediate corollary of the Fourier calculation, not a claim of a new classical-constant evaluation.

5.2 Exponentially convergent numerical reproduction

For $t > 0$, put

$$C(t) = \text{Li}_1(e^{-t}) = -\log(1 - e^{-t}), \quad F(t) = \partial_s \text{Li}_s(e^{-t}) \Big|_{s=1}.$$

The usual Mellin representation of the Tornheim sum gives

$$T_{12} = \int_0^\infty F(t)^2 dt, \quad T_{13} = \int_0^\infty F(t)C(t)(\log t + \gamma) dt, \quad T_3 = \int_0^\infty C(t)^2(\log t + \gamma) dt. \quad (5.4)$$

For $0 < t < 2\pi$, cancellation of the two poles in Jonquière's expansion produces

$$\begin{aligned} C(t) &= -\log t + \sum_{k \geq 1} c_k t^k, & c_k &= \frac{(-1)^k \zeta(1-k)}{k!}, \\ F(t) &= -\alpha - \gamma \log t - \frac{1}{2} \log^2 t + \sum_{k \geq 1} b_k t^k, & b_k &= \frac{(-1)^k \zeta'(1-k)}{k!}, \\ \alpha &= \gamma_1 + \frac{\gamma^2}{2} + \frac{\pi^2}{12}. \end{aligned}$$

Here the Stieltjes convention is $\zeta(1+z) = z^{-1} + \sum_{j \geq 0} (-1)^j \gamma_j z^j / j!$. For computational stability the coefficient formulas can be rewritten as

$$\begin{aligned} c_1 &= \frac{1}{2}, & b_1 &= \frac{1}{2} \log(2\pi), \\ c_{2j} &= \frac{(-1)^j \zeta(2j)}{j(2\pi)^{2j}}, & b_{2j} &= c_{2j} \left(\log(2\pi) - \psi(2j) - \frac{\zeta'(2j)}{\zeta(2j)} \right), \\ c_{2j+1} &= 0 \quad (j \geq 1), & b_{2j+1} &= \frac{(-1)^{j+1} \pi \zeta(2j+1)}{(2j+1)(2\pi)^{2j+1}} \quad (j \geq 1). \end{aligned}$$

The functional equation of the Riemann zeta function proves these identities. Elementary bounds $\zeta(k) < 2$, $|\zeta'(k)| < 2$, $H_{k-1} \leq k-1$, $\gamma < 1$, $\log(2\pi) < 2$, and $2\pi > 6$ give, for $k \geq 2$,

$$|b_k| \leq 12 \cdot 6^{-k}, \quad |c_k| \leq 2 \cdot 6^{-k}.$$

Thus after degree $K \geq 1$ their remainders on $0 < t \leq 1$ are at most

$$\epsilon_b = \frac{12}{5} 6^{-K}, \quad \epsilon_c = \frac{2}{5} 6^{-K}.$$

All integrals of the resulting polynomials in t and $\log t$ are evaluated using $\int_0^1 t^k \log^j t dt = (-1)^j j! / (k+1)^{j+1}$.

On $t \geq 1$, insert $C(t) = \sum_{n \geq 1} e^{-nt}/n$ and $F(t) = -\sum_{n \geq 1} (\log n) e^{-nt}/n$. The integral with the logarithmic factor uses

$$\int_1^\infty e^{-kt} (\log t + \gamma) dt = \frac{E_1(k) + \gamma e^{-k}}{k}.$$

Truncate this sum at $m+n \leq N$. A common upper bound for the three omitted tails is

$$B_N = 4 \sum_{k > N} k e^{-k} = \frac{4e^{-(N+1)}((N+1) - Ne^{-1})}{(1 - e^{-1})^2}.$$

For the three integrals in the order T_{12}, T_{13}, T_3 , safe total series-truncation bounds are respectively

$$\epsilon_b(12 + \epsilon_b) + B_N, \quad 5\epsilon_b + 15\epsilon_c + 2\epsilon_b\epsilon_c + B_N, \quad 10\epsilon_c + 2\epsilon_c^2 + B_N.$$

For example the elementary estimates $C(t) \leq -\log t + 1$ and $|F(t)| \leq 4 - \log t + \frac{1}{2} \log^2 t$ on $(0, 1]$ imply all the constants above. The sharper bound $\int_0^1 |F(t)| dt < 6$ follows directly from its

coefficient expansion (one may use $0 < \alpha < 2$). An elementary verification of the latter constant bound is available: the defining limit for γ_1 and the total variation of $x \mapsto \log x/x$ on $[1, \infty)$ give $|\gamma_1| \leq 2/e < 3/4$; combine this with $0 < \gamma < 3/5$ and $3 < \pi < \sqrt{12}$. This is an analytic truncation certificate; floating-point rounding must be handled separately to turn an implementation into interval arithmetic.

The independent implementation `verify_tornheim.py` evaluates the three partials and compares (5.1) against direct log-gamma quadrature. Its output explicitly labels the rounding qualification.

5.3 A sign check for an auxiliary identity

If the June 2012 preprint is used, an auxiliary sign should be checked before copying its Remark 3. Put

$$S_\gamma = \sum_{m,n \geq 1} \frac{(\gamma + \log m)(\gamma + \log n)}{mn(m+n)}.$$

Directly expanding the numerator, with absolutely convergent sums, gives

$$T_{12} = S_\gamma + 2\gamma(Z'(2) + \zeta'(3)) - 2\gamma^2\zeta(3).$$

The coefficient of S_γ is positive. The displayed minus sign in equations (63) and (65) of the author-hosted June 2012 preprint would make their right side negative, although $T_{12} > 0$. This issue does not affect the cubic formula proved above. This note does not assert that the same typographical error occurs in the final journal typesetting.

6 Corrections, integration notes, and further research

6.1 Proposed corrections to the pinned manuscript

The following corrections are scoped to the pinned source, so that a maintainer can distinguish a stale manuscript statement from a gap in the repository as a whole. Some relevant identities are already present in existing continuation material or its exact coefficient code. The proposed patch updates a few local statements; this article supplies the fuller proofs and context.

C1. The cubic moment has a named evaluation. Chapter 7, at `integral:neg:cubic`, calls the identification of a named depth-two atom an open problem. Bailey–Borwein–Borwein [4, Theorem 6] already provide a Tornheim–Witten derivative evaluation. Replace that description by the formula in Section 5, or a reference to it, and state the narrower remaining problem: reduction of the prescribed derivative combination to a specifically defined smaller basket. The cubic-moment discovery paragraph in Chapter 10 should make the same distinction. A failed numerical search in a smaller basket establishes no nonmembership statement.

C2. Two weight-five coordinates suffice. The paragraph before `gauss:eq:S4-closed` in Chapter 4 infers only an at-most-three spanning statement from its sporadic relation. Proposition 4.1 supplies two independent exact shuffle rows, so at most two g -coordinates remain over the displayed single products. The sporadic relation is a proved consequence. Its proof status and its rank consequence should both be updated, while the mixed S_4 equation itself remains conjectural.

C3. The imaginary weight-six mixed survivors disappear. The statements around `mixed:sec:eisdim` and `mixed:thm:parallel` propose imaginary weight-six mixed directions at both levels. Equations (4.7) and (4.8) give explicit reductions. The all-weight inverse-argument coefficient function in the existing report-10 code `10-exact-structure-verify_mixed_and_euler.py` already contains the relevant reduction mechanism. These manuscript survivor counts should therefore be revised. The real weight-five question is unaffected by this argument and should be treated separately.

C4. A coefficient label in the sixth-point proof. In Chapter 7’s proof of `stieltjes:thm:G1half`, the text describes $\log^2 6 - \log^2 3 - \log^2 2$ as a coefficient of $\zeta''(0)$. It is the coefficient of $\zeta(0)$. Indeed, for $P(s) = 6^s - 3^s - 2^s + 1$,

$$P(0) = P'(0) = 0, \quad (P\zeta)''(0) = P''(0)\zeta(0) = (\log^2 6 - \log^2 3 - \log^2 2)\zeta(0).$$

The displayed final identity in that proof is unchanged.

C5. Scope of dimension and independence language. All matrix-rank statements should specify the formal space, background basket, convergence or regularization rules, and exact row family. Theorems 4.4 and 4.5 provide such a specification for one imaginary quotient. They do not validate claims about the dimensions of all actual periods. Likewise, non-reduction in a bounded-height search should be recorded as a search result with the basket and bounds, not as a theorem of arithmetic independence. The existing manuscript often gives this warning; the surviving stronger wording should be made consistent.

C6. An auxiliary sign in a cited preprint. The June 2012 author-hosted version of [4], equations (63) and (65), prints a minus sign before a manifestly positive quadratic sum. Section 5 gives a direct sign check and the corrected plus sign. This observation is restricted to the inspected preprint. It should not be generalized to an uninspected journal version.

6.2 Integration and claim status

A practical integration sequence is to update the short correction paragraphs first, then add the quantitative moment and Herglotz sections as separate research continuations, and finally replace finite experimental dimension summaries by explicitly scoped theorems where applicable. The selected S_6 equation belongs in the conjecture ledger with its frozen integer vector and exact residual certificate.

The machine-readable `claim_ledger.json` distinguishes analytic theorems, exact formal-algebra theorems, classical rederivations, source corrections, and unresolved conjectures. The package excludes exploratory rejected candidates and does not turn a small residual into an accepted identity. All paths needed to rebuild the article are relative, and no repository-specific absolute path is required by the LaTeX source.

6.3 A proposed research programme

The following questions have concrete starting points and distinct success criteria. Their order reflects likely usefulness rather than a claim that any is easy.

R1. Prove the two even-index harmonic reductions. Find a functional or iterated-integral relation proving Conjecture 4.6 and then Conjecture 4.7. The separating functional shows that

re-solving the same single-product matrix cannot suffice. Candidate routes include relations lifted through higher depth and the extra symmetries of the level-four punctured sphere. A useful certificate would give the new relation explicitly and show that its image under ℓ is nonzero before eliminating it with the other rows. Zhao [9] supplies relevant broader context, but the current obstruction does not prove that any one of those routes is necessary.

- R2. Determine an even-index family with a justified basket.** For S_{2m} , decide which g -coordinates and single products are actually sufficient at each weight. The S_4 and S_6 equations suggest structured reductions, but two examples do not determine a general coefficient law. Choose the basket from exact relations, freeze it, record a height bound before each search, and validate any candidate with the rational Euler method. The desired outcome is a uniform formula or a precise counterexample to a proposed family, rather than an unqualified claim that odd weights require new constants.
- R3. Extend the formal quotient to other levels.** Compute the analogue of $V_{L,w}$ first for $L = 5$ and 6, with exactly the same row conventions. Determine whether its polynomial description comes from a finite group, a module over an invariant ring, or a more complicated relation. An all-weight Hilbert series should be accompanied by explicit spanning and separating families, not inferred from a short table.
- R4. Compare restricted and larger relation presentations.** Add depth-two distribution, weight-one relations, and lifted higher-depth rows in a documented order. At each step compute the kernel induced on the polynomial parameters of Theorems 4.4 and 4.5. This would show exactly which new rows remove which obstructions. A later comparison with motivic bounds must keep formal dimension, period dimension, and conditional period conjectures distinct.
- R5. Uniform moment truncation beyond the least-term window.** Theorem 2.5 treats bounded $\eta = N\Lambda - n$. Develop a uniform expansion when N/n varies over a positive interval, and then when it tends to zero or infinity. The inverse remainder kernel and gamma concentration suggest the relevant starting formula. The challenge is to retain uniform error estimates through the changing concentration region. A second target is a fully explicit finite- n, N remainder bound.
- R6. Global analytic structure of the inverse gamma branch.** Locate the next singularities reached by analytic continuation of the particular branch g , including which critical values actually lie on that branch. Determine their contributions after the leading square-root expansion is removed. The local slit-disk argument used here deliberately avoids an unsupported assertion about a global inverse. Understanding the further singularities could explain exponentially smaller corrections to the late coefficients and constrain possible global sign bounds.
- R7. Higher Herglotz corrections and transition in the argument.** Compute several more half-integer terms in the oscillatory expansion and give explicit error constants. More substantially, analyze simultaneous limits such as $x/r \rightarrow \xi > 0$ or $x \rightarrow 0$ with r . The fixed- x , compact-uniform theorem does not automatically cover these regimes: the optimal Mellin contour and the contributing arithmetic sectors change. A useful result would identify the transition from one dominant sector to a collective contribution.
- R8. Quantitative Herglotz sign locations.** Theorem 3.4 gives infinitely many signs, but does not identify every sign for every large integer r . Use higher corrections and explicit bounds to

locate the changes near the zeros of the leading cosine. Because the phase increment tends to zero, an additive error estimate is essential near these zeros. The target is an eventually certified sign algorithm and an asymptotic count of transitions.

- R9. Arithmetic twists of the high-derivative theorem.** Replace the divisor coefficients by character-twisted coefficients or consider higher Herglotz functions. The functional-equation framework of [5, 6] is a natural starting point. Determine how parity, conductor, and vanishing of the first arithmetic coefficient alter the exponential rate and oscillatory phase. The sum of all sectors must again be bounded uniformly; a single special-function asymptotic is not enough.
- R10. Minimal derivative coordinates for the cubic moment.** The cubic formula singles out $2T_{13} - T_{12}$, together with $T_3 = 2Z'(2)$. Determine the exact relations among Tornheim derivatives at $(1, 1, 1)$ obtained from symmetry, integration by parts, and differentiated functional equations. State a candidate basis before asking whether this combination reduces to ordinary zeta derivatives. Failure of such a reduction search should not be confused with the already solved problem of finding a named depth-two representation.
- R11. Certified evaluation of the asymptotic quantities.** Combine the explicit tail estimates with interval implementations of gamma, polygamma, Hurwitz zeta, and the remaining finite integrals. For moments, the cutoff identity (2.17) provides a convergent route; for Herglotz remainders, (3.1) provides a simple arithmetic cutoff. The rational S_6 certificate is a model of the distinction between a proved truncation bound and an actual certified implementation.
- R12. Repository-level mathematical provenance.** Attach to every accepted identity its exact normalization, definition of the basis, proof source, executable check, and status. For dimension claims also store the row generator and a basis or separating witness. For numerical conjectures store the frozen coefficients separately from discovery logs, with an independent holdout computation. This would prevent a correction already proved in a continuation from remaining an open claim in the consolidated manuscript.

These questions connect the analytic and algebraic sides of the programme: good coordinates make strong conjectures visible, while controlled remainders and explicit relation spaces show exactly what has, and has not, been proved.

7 Reproducibility, numerical evidence, and limitations

The package includes the complete LaTeX source, generated figures, exact relation-space code, numerical diagnostics, rational certificates, a claim ledger, and a proposed source patch. Computations are separated into exact arithmetic and ordinary high-precision arithmetic. The analytic proofs are in the article; the numerical checks provide independent diagnostics for constants, signs, branches, and truncation scales.

7.1 Moment coefficients and least-term errors

The recurrence in Section 2 generates b_k without numerically inverting gamma. The script `verify_moments.py` also checks the first eight coefficient polynomials against a separate formal

exponential-series expansion. For direct moments it substitutes $x = e^{-t}$ and integrates

$$m_n = \int_0^\infty \exp(n \log(\log \Gamma(e^{-t})) - t - \log \Gamma(n+1)) dt.$$

The upper limit is infinity. A fixed short cutoff may approximate m_n accurately while corrupting the far smaller least-term error, so it is unsuitable for this check unless its tail is explicitly controlled.

The computations use 180 decimal working digits. Orders 20 and 80 are repeated at 215 digits; the respective differences between normalized remainder ratios are below 3×10^{-162} and 6×10^{-67} . The latter loses many digits through the deliberate subtraction of a remainder of order 10^{-113} ; the extra precision is therefore material.

n	N	$\eta = N\Lambda - n$	error / first omitted term	N^3 (error after two corrections)
10	9	1.388939	0.503338844959	0.059202
20	17	1.512440	0.499910660977	0.056121
30	25	1.635941	0.498697774004	0.055539
40	33	1.759442	0.498075743812	0.052612
60	49	2.006443	0.497442625716	0.042058
80	64	0.988008	0.501995818744	0.046583

The final column refers to the normalized ratio in (2.16), not to M_n itself. The bounded values corroborate the predicted $O(N^{-3})$ term; they are not used to prove it.

7.2 Herglotz derivatives

The finite rational formula is compared against a separate positive real integral at eight rational arguments and five derivative orders, giving 40 comparisons. The maximum observed residual at 70 working digits is about 1.27×10^{-70} . The coefficient construction uses exact SymPy Bernoulli polynomials; the numerical integral uses independently generated Eulerian coefficients for Li_{1-r} .

Seven higher-order checks, at $r = 10, 20, 30, 50, 80, 100, 150$, increase working precision with the size of the Bernoulli sum and repeat at 40 additional digits. This avoids mistaking cancellation error for the exponentially small E_r . Selected values at $x = 1$ are

r	$E_r(1)$
20	$9.5700353115338104 \times 10^{-8}$
30	$-1.8039764729798573 \times 10^{-9}$
50	$-4.2383422584491617 \times 10^{-12}$
100	$6.4956183476241665 \times 10^{-17}$
150	$-1.2668021540982769 \times 10^{-20}$

Three finite divisor- U sums are compared with the exact rational derivative formula and both available analytic tail bounds. Twelve checks of the single-sector saddle expansion cover $x = 1/2, 1, 2$ and four derivative orders. The figure uses 57 further exact-formula evaluations through $r = 300$, with precision chosen to cover cancellation. These checks confirm the implementation and the phase normalization; the sign theorem is analytic.

7.3 Cyclotomic relation spaces and the conjecture certificate

The exact row generator checks both levels through weight 16: 30 rational rank calculations, together with verification that the proposed polynomial bases annihilate every row and have full predicted

rank. The weight-five separating functional is checked entry by entry. The all-weight proofs in the article establish the dimensions beyond that finite range.

Separate 90-digit integral diagnostics verify the two weight-five shuffle rows, both versions of S_4 , its equivalent $A_{1,4}$ form, and the inverse-argument weight-six identities. Residuals for the two S_4 forms are about 1.53×10^{-92} . This is new corroboration of an existing conjecture, not its proof.

The S_6 evidence has three independent layers: a 220-digit Mellin-integral computation, a 300-digit Euler-series computation, and an exact rational enclosure. The last uses only Python's standard library and the proved tails in Theorem 4.8. It is therefore a certificate of the bound (4.20), including rounding, rather than a heuristic precision report. The residual interval contains zero; it cannot distinguish an exact identity from a sufficiently small nonzero value.

7.4 Reproduction of the classical cubic evaluation

Using small- t degree 120, large- t diagonal cutoff 240, and 80 working digits, the Tornheim implementation gives

$$\begin{aligned} T_3 &= -4.85147944680912469492172639886220773949\dots, \\ T_{12} &= 4.30244762034222643331985179818698942752\dots, \\ T_{13} &= 8.71447281121431423816300785446242065162\dots, \\ M_3 &= 5.74038880722947428001957168810246146296\dots \end{aligned}$$

The direct moment integral and the Tornheim expression differ by about 1.69×10^{-80} . The three analytic series-truncation bounds are respectively below 1.3×10^{-92} , 7.6×10^{-93} , and 1.7×10^{-93} . These bounds control series truncation, while the implementation's floating-point rounding and quadrature remain separately qualified. The mismatch between the sizes of the truncation bounds and the observed residual is expected at 80-digit working precision.

7.5 Running the package

The root `README.md` and `Makefile` give the complete commands. A typical sequence is

```
python3 -m pip install -r requirements.txt
make exact
make numeric
make figures
make pdf
```

The exact S_6 certificate alone requires no third-party package:

```
python3 scripts/certify_s6.py \
  --output data/s6_rational_certificate.json
```

The PDF build uses ordinary `pdflatex/latexmk` and standard TeX packages; its input figures are included. All output data needed to inspect the delivered computations are already present, so rerunning every numerical calculation is optional.

A source manifest pins the research baseline, and a SHA-256 manifest covers the delivered artifacts. The correction patch is a proposal for the pinned source; it should be reviewed against any newer upstream revision before application. No external service is needed to run the certificate, relation-space checks, or article build.

References

- [1] V. Reshetnikov and contributors, *ProveIt: Polylogarithms manuscript and research continuations*. Source baseline: commit 78f9e5df05eb0abb16c89fb00b9e89403598bd17, 9 October 2026. <https://github.com/VladimirReshetnikov/ProveIt/tree/78f9e5df05eb0abb16c89fb00b9e89403598bd17/Analysis/Polylogarithms/docs>. Exact examined paths and content hashes are recorded in `source_manifest.json`.
- [2] T. Amdeberhan, M. W. Coffey, O. Espinosa, C. Koutschan, D. V. Manna, and V. H. Moll, *Integrals of powers of loggamma*, Proc. Amer. Math. Soc. **139** (2011), no. 2, 535–545. doi:10.1090/S0002-9939-2010-10589-0. Author preprint: https://www.math.tulane.edu/~vhm/papers_html/lg-subm.pdf. Section 8 gives the classical moment expansion.
- [3] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009. The smooth implicit-function schema and singularity transfer are the classical background for the late-coefficient analysis.
- [4] D. H. Bailey, D. Borwein, and J. M. Borwein, *On Eulerian log-gamma integrals and Tornheim–Witten zeta functions*, Ramanujan J. **36** (2015), 43–68. doi:10.1007/s11139-012-9427-1. Author-hosted preprint dated 27 June 2012: <https://carmamaths.org/resources/jon/log-gamma.pdf>. Theorem 6, equation (61), is the cubic evaluation discussed here. The auxiliary sign observation is scoped to this preprint’s equations (63) and (65).
- [5] D. Radchenko and D. Zagier, *Arithmetic properties of the Herglotz function*, J. Reine Angew. Math. **797** (2023), 229–253. doi:10.1515/crelle-2023-0009. Author preprint: <https://people.mpim-bonn.mpg.de/zagier/files/preprints/HerglotzFunction.pdf>; arXiv:2012.15805. See equation (6), the paragraph after Proposition 4, and Section 7.3.
- [6] A. Dixit, S. Sathyanarayana, and N. Guru Sharan, *Mordell–Tornheim zeta functions and functional equations for Herglotz–Zagier type functions*, Adv. Math. **473** (2025), article 110303. doi:10.1016/j.aim.2025.110303. Author preprint: <https://adixit.people.iitgn.ac.in/wp/wp-content/uploads/2024/05/dss2.pdf>; arXiv:2405.07934. The literature comparison in this continuation uses the accessible author-hosted preprint.
- [7] N. M. Temme, *Uniform asymptotic expansions of a class of integrals in terms of modified Bessel functions, with application to confluent hypergeometric functions*, SIAM J. Math. Anal. **21** (1990), no. 1, 241–261. doi:10.1137/0521013.
- [8] National Institute of Standards and Technology, *NIST Digital Library of Mathematical Functions*, Section 13.4, equation 13.4.4, and Section 13.8(iii), equations 13.8.11–16. <https://dlmf.nist.gov/13.4.E4>; <https://dlmf.nist.gov/13.8.iii>. Consulted 9–10 October 2026.
- [9] J. Zhao, *Multiple polylogarithm values at roots of unity*, C. R. Acad. Sci. Paris, Ser. I **346** (2008), 1029–1032. doi:10.1016/j.crma.2008.09.011. <https://www.numdam.org/item/10.1016/j.crma.2008.09.011.pdf>. Its standard-relation framework includes more relations and depths than the restricted presentation computed in this article.